Erdős Problem #287: A Standalone Verification Dossier

Draft V12 – explicit Ford certificate, smooth-parity counterguard, and corrected fixed-leakage frontier

Research draft – not a solution claim

27 August 2026

Purpose and status. This document is deliberately written so that the mathematical chain can be audited from this document alone. It does not use "see the Lean bank", "see the Gate-1A paper", or "see the Gate-0/2 archive" as substitutes for mathematical statements. The finite Erdős-287 reduction is proved explicitly below. The Ford–Maynard sequence conditions are written out explicitly. Gate 1A and Gate 1B are reproduced as mathematical theorem chains, with every load-bearing hypothesis stated. Gate 2 is written as an explicit conditional endgame. Where a statement remains unproved, the statement itself is displayed rather than hidden behind a status label. No claim of Full Type II, twin-prime infinitude, or a solution of Problem #287 is made.

Abstract

Let A be a finite set of distinct integers $1 < n_1 < \cdots < n_k$ with $\sum_{a\in A} 1/a = 1$. Erdős Problem #287 asks whether some consecutive gap must be at least three. We first give a self-contained finite reduction. The reciprocal identity implies a top-layer p-adic congruence. Together with the elementary half-range bound $N \le \lfloor M/2 \rfloor$, this shows that a prime $q > 3$ with $3q > M$ and $q^2 > M$ forces both q and $2q$ to be absent, while any prime $p \in (M/2, M]$ is absent. Hence an interior prime pair $p = 2q \pm 1$ produces two adjacent holes and contradicts the gap-≤ 2 hypothesis. A fixed band $q \in (3X/4, 4X/5)$ with $X = \lfloor M/2 \rfloor$ suffices for every $M \ge 9$.

The analytic task is therefore to force such a Sophie-type witness. We define the comparison density

$$B(n) = \mathfrak{S}_2 \prod_{\ell | n, \ell > 2} \frac{\ell - 1}{\ell - 2},$$

and prove $Y^{-1} \sum_{m \le Y} B(dm) \to d/\varphi(2d)$, which is the exact local mean required by the two affine forms $2n \pm 1$. We then write out the Ford–Maynard Type-I, Type-II, growth, comparison and bounded-class hypotheses, the 1/6 parameter geometry, the Gate-0 specialization, the complete current Gate-1A conditional mechanism, the complete current Gate-1B conductor reduction and its late pair-modulus reorganisation, the source-exhaustive Full-Type-II socket, and the Gate-2 rough endgame. Draft V12 preserves the corrected Ford–Maynard N_2 replay from Draft V8 and incorporates the 26/27 August Gate-1B and Ford-side updates: H7 short-short now has a conditional joint-prime compiler with relative capacity $Y^{-1/2} = X^{-1/18}$; the H8 analogue has the same research-level fixed-power capacity but still lacks its formal source/robustness pin; the QK56 full-covariance parent is promoted internally to an analytic pass modulo literal source pins, while the principal Gate-1B analytic candidate moves to the balanced shifted-quotient child after a mandatory literal TT^* source recovery. The canonical Gate-1B additive zero mode remains an exact algebraic pass. Downstream, Ford–Maynard five-collar geometry now gives a source-specific organisation of the rough N_2 sector, the grouped equation (7.23) remains the proof-specific Type-II landing point, and the first unresolved Gate-to-Ford interface is the literal weighted packet dictionary labelled SOURCE-ADAPTER45. A subsequent hostile census corrects the first interpretation of this fixed-certificate pivot. Ford–Maynard supply an explicit certificate g_0 at their central point; after a small fixed

shrink one obtains a literal certificate g_* and arithmetic weight H_*. The fixed-certificate transference identity is valid, but the resulting leakage family does *not* collapse to only H8/H9 and QK56 packets. Already the $k = 0$, $J = \varnothing$ cell contains an exact smooth-parity packet with the truncated Möbius divisor detector $\sum_{d|n,\, d\le n^{1/2-\varepsilon_*}} \mu(d)$. Balanced squarefree cells show nonzero contributions at orders $7, 8, 9, 10, 11, 12, \ldots$, so an "H8/H9-only" census is false. The generic seven-prime quadratic-Kummer estimate remains a useful conditional child provider with relative capacity $X^{-1/18+o(1)}$, but it is no longer the controlling first wall. The first literal analytic open in the fixed-certificate route is the source-exact smooth-parity packet `287-FIXED-CERTIFICATE-SMOOTH-PARITY45`; the full fixed-certificate leakage estimate and Problem #287 remain open.

Contents

Part I

Erdős #287: the finite reduction written from first principles

1 Problem, notation, and the hypothetical counterexample

Let

$$A = \{n_1 < \cdots < n_k\} \subset \mathbb{N}, \qquad n_1 > 1, \qquad \sum_{i=1}^{k} \frac{1}{n_i} = 1.$$

Write

$$N := n_1, \qquad M := n_k.$$

Assume for contradiction that

$$n_{i+1} - n_i \le 2 \qquad (1 \le i < k). \tag{G2}$$

Call an integer $h \in [N, M]$ a *hole* if $h \notin A$.

Lemma 1.1 (Two adjacent holes are impossible)**.** *Under* (G2)*, there do not exist $x, x+1 \in [N, M]$ which are both holes.*

Proof. Since $N, M \in A$, neither endpoint can be a hole. Thus if $x, x+1$ are holes there is a largest denominator $a \in A$ with $a < x$ and a smallest denominator $b \in A$ with $b > x+1$. Then $b - a \ge 3$, contradicting (G2). □

2 The half-range placement bound

Lemma 2.1 (Half-range placement)**.** *Every hypothetical counterexample satisfies*

$$N \le \left\lfloor \frac{M}{2} \right\rfloor.$$

Proof. Suppose instead $N \ge \lfloor M/2 \rfloor + 1$. Then $2N \ge M + 1$, hence

$$k \le M - N + 1 \le N.$$

Because the denominators are distinct and at least N,

$$\sum_{i=1}^{k} \frac{1}{n_i} \le \frac{1}{N} + \frac{k-1}{N+1} < \frac{k}{N} \le 1,$$

contradicting the reciprocal identity. (If $k = 1$, the sole denominator would be 1, excluded by hypothesis.) □

3 The top-layer congruence derived directly from the reciprocal identity

Theorem 3.1 (Top-layer congruence)**.** *Fix a prime p. Let*

$$E := \max_{a \in A} v_p(a), \qquad I_p := \{a \in A : v_p(a) = E\}.$$

For each $a \in I_p$ write $a = p^E m_a$ with $p \nmid m_a$. Then

$$\boxed{\sum_{a \in I_p} m_a^{-1} \equiv 0 \pmod{p}.} \tag{TL}$$

Here m_a^{-1} is the inverse of m_a modulo p.

Proof. Let $L = \operatorname{lcm}(A)$ and write $L = p^E L_0$ with $p \nmid L_0$. Multiplying $\sum_{a \in A} 1/a = 1$ by L gives

$$L = \sum_{a \in A} \frac{L}{a}.$$

Reduce modulo p. If $v_p(a) < E$, then L/a is divisible by p, so these terms vanish. If $a = p^E m_a \in I_p$, then

$$\frac{L}{a} = \frac{L_0}{m_a} \equiv L_0 m_a^{-1} \pmod{p}.$$

Since $L \equiv 0 \pmod{p}$ and L_0 is a unit modulo p,

$$0 \equiv L_0 \sum_{a \in I_p} m_a^{-1} \pmod{p},$$

which proves (TL). □

4 Immediate prime exclusions

Lemma 4.1 (A two-multiple prime fibre is empty)**.** *Let $q > 3$ be prime and suppose*

$$3q > M, \qquad q^2 > M.$$

Then neither q nor $2q$ belongs to A.

Proof. The only multiples of q in $[1, M]$ are q and $2q$, and $q^2 > M$ ensures both have q-adic valuation one. If at least one belonged to A, the maximal q-adic fibre would be one of

$$\{q\}, \qquad \{2q\}, \qquad \{q, 2q\}.$$

By Theorem 3.1, these would force respectively

$$1 \equiv 0, \qquad 2^{-1} \equiv 0, \qquad 1 + 2^{-1} = \frac{3}{2} \equiv 0 \pmod{q},$$

all impossible for $q > 3$. □

Lemma 4.2 (Every top-half prime is a hole)**.** *If p is prime and*

$$\frac{M}{2} < p \leq M,$$

then $p \notin A$.

Proof. The only multiple of p in $[1, M]$ is p. If $p \in A$, the top p-adic fibre is the singleton $\{p\}$, and (TL) gives $1 \equiv 0 \pmod{p}$. □

5 The direct $\pm$ Sophie blockers

Proposition 5.1 (Plus-Sophie blocker)**.** *Let $q > 3$ and $p = 2q + 1$ be primes. If*

$$p \le M < 3q, \qquad q^2 > M,$$

then the assumed counterexample cannot exist.

Proof. Lemma 4.1 makes $2q = p - 1$ a hole. Since

$$M < 3q < 4q + 2 = 2p,$$

we have $p > M/2$, so Lemma 4.2 makes p a hole. Lemma 2.1 gives $N \le \lfloor M/2 \rfloor < p - 1$, and $p \le M$. Hence $p - 1, p$ are adjacent holes in $[N, M]$, contradicting Lemma 1.1. $\square$

Proposition 5.2 (Minus-Sophie blocker)**.** *Let $q > 3$ and $p = 2q - 1$ be primes. If*

$$p + 1 \le M < 3q, \qquad q^2 > M,$$

then the assumed counterexample cannot exist.

Proof. Lemma 4.1 makes $2q = p + 1$ a hole. Since $q > 3$,

$$M < 3q < 4q - 2 = 2p,$$

so $p > M/2$ and Lemma 4.2 makes p a hole. Lemma 2.1 gives $N \le \lfloor M/2 \rfloor < p$, while $p + 1 \le M$. Hence $p, p + 1$ are adjacent holes, contradiction. $\square$

6 A fixed analytic band

Put

$$X := \left\lfloor \frac{M}{2} \right\rfloor$$

and fix a non-negative $W \in C_c^\infty((3/4, 4/5))$.

Lemma 6.1 (Band arithmetic)**.** *For every integer $M \ge 9$, if $W(q/X) > 0$, then*

$$\boxed{q > 3, \qquad M < 3q, \qquad 2q + 1 \le M, \qquad q^2 > M.}$$

Proof. We have $3X/4 < q < 4X/5$ and $X = \lfloor M/2 \rfloor$. Since $M \ge 9$, $X \ge 4$, hence $q > 3X/4 \ge 3$, and the strict inequality gives $q > 3$.

For the lower Sophie-band inequality, $X \ge (M - 1)/2$, so

$$3q > \frac{9X}{4} \ge \frac{9(M-1)}{8} \ge M,$$

where the last inequality is equivalent to $M \ge 9$. Also $q < 4X/5 \le 2M/5$, hence

$$2q + 1 < \frac{4M}{5} + 1 \le M \qquad (M \ge 5).$$

Finally, strictness is important at the endpoint. From

$$q > \frac{3X}{4} \ge \frac{3(M-1)}{8}$$

and

$$\left(\frac{3(M-1)}{8}\right)^2 - M = \frac{9M^2-82M+9}{64} = \frac{9(M-9)(M-1/9)}{64} \geq 0$$

for $M \geq 9$, we obtain

$$q^2 > \left(\frac{3(M-1)}{8}\right)^2 \geq M.$$

Thus the threshold $M \geq 9$ is valid; no replacement by $M \geq 10$ is required. □

Corollary 6.2 (Finite landing theorem)**.** *For $M \geq 9$, if there exists a prime q with $W(q/X) > 0$ such that at least one of $2q-1, 2q+1$ is prime, then no counterexample with maximum denominator M exists.*

7 The exact comparison density for the two affine prime forms

Define

$$\mathfrak{S}_2 := \prod_{\ell>2} \frac{\ell(\ell-2)}{(\ell-1)^2}, \qquad B(n) := \mathfrak{S}_2 \prod_{\substack{\ell \mid n \\ \ell>2}} \frac{\ell-1}{\ell-2}.$$

Let

$$a_X(n) := W(n/X)\big(\Lambda(2n-1)+\Lambda(2n+1)\big),$$
$$b_X(n) := 4W(n/X)B(n), \qquad w_X(n) := a_X(n) - b_X(n).$$

For odd squarefree e set

$$g(e) := \prod_{\ell \mid e} \frac{1}{\ell-2},$$

and put $g(e) = 0$ otherwise. Then

$$B(n) = \mathfrak{S}_2 \sum_{e \mid n} g(e). \tag{7.1}$$

Proposition 7.1 (Local mean)**.** *For each fixed positive integer d,*

$$\lim_{Y\to\infty} \frac{1}{Y} \sum_{m \leq Y} B(dm) = \frac{d}{\varphi(2d)}.$$

Proof. The series

$$\sum_e \frac{|g(e)|}{e} = \prod_{\ell>2}\left(1+\frac{1}{\ell(\ell-2)}\right)$$

converges absolutely. Truncate at $e \leq Z$, average, and then let $Y \to \infty$ followed by $Z \to \infty$. Since the density of m with $e \mid dm$ is $(e,d)/e$,

$$\lim_{Y\to\infty} \frac{1}{Y} \sum_{m \leq Y} B(dm) = \mathfrak{S}_2 \sum_e g(e)\frac{(e,d)}{e}.$$

For an odd prime $\ell \nmid d$, the Euler factor is

$$\frac{\ell(\ell-2)}{(\ell-1)^2}\left(1+\frac{1}{\ell(\ell-2)}\right) = 1.$$

For $\ell \mid d$, it is

$$\frac{\ell(\ell-2)}{(\ell-1)^2}\left(1+\frac{1}{\ell-2}\right)=\frac{\ell}{\ell-1}.$$

The prime 2 contributes no factor. Hence the product is $\prod_{\ell\mid d,\ell>2}\ell/(\ell-1)=d/\varphi(2d)$. □

Remark 7.2 (Why the factor 4 is exact). For each sign, the affine form $2dm \pm 1$ has von Mangoldt mean $2d/\varphi(2d)$, so the two signs together have mean $4d/\varphi(2d)$, exactly matching $4B(dm)$.

Part II

The Ford–Maynard framework and Gate 0, written explicitly

8 The exact Ford–Maynard sequence class

This section writes out the external sieve framework used downstream. The notation follows Ford–Maynard [3], but the formulas needed here are reproduced so that the application can be audited locally. Fix parameters

$$0<\gamma<1, \qquad 0\le\theta<\frac{1}{2}, \qquad 0<\nu\le 1-\theta,$$

and non-negative sequences $(a_n), (b_n)$ supported on $x/2<n\le x$. Put $w_n = a_n - b_n$.

8.1 Type I, Type II, and the growth condition

The Type-I hypothesis is

$$\boxed{\sum_{m\le x^\gamma}\tau^B(m)\max_{I\text{ interval}}\left|\sum_{\substack{x/2<mn\le x\\ n\in I}} w_{mn}\right|\le\frac{x}{(\log x)^B}.} \tag{I}$$

The Type-II hypothesis is universal in the coefficients:

$$\boxed{\left|\sum_{\substack{(x/2)^\theta<m\le x^{\theta+\nu}\\ x/2<mn\le x}}\xi_m\kappa_n w_{mn}\right|\le\frac{x}{(\log x)^B}} \tag{II}$$

for every pair of complex sequences satisfying $|\xi_m|\le\tau^B(m)$ and $|\kappa_n|\le\tau^B(n)$. The lower endpoint is $(x/2)^\theta$, not x^θ.

The ordinary growth condition is

$$\boxed{\sum_n |w_n|\tau(n)\le x(\log x)^\varpi, \qquad w_n\ge -x^{\nu/10}\quad (x/2<n\le x).} \tag{w}$$

8.2 The comparison-sequence conditions

Let $P = (\gamma, \theta, \nu)$. Define $R(P)$ to be the set of finite vectors $u = (u_1, \ldots, u_k)$ whose components lie in $(0, 1-\gamma)$, sum to 1, and have no proper subsum in $[\theta, \theta+\nu]$. Let $C(R(P))$ denote all coagulations of vectors in $R(P)$.

The first comparison condition is the prime-mass lower bound

$$\boxed{\sum_p b_p \geq \frac{x}{(\log x)^{\varpi}}.} \tag{b.1}$$

The second is a generalized prime-number-theorem condition. For every $2 \leq k \leq 1/\nu$, every convex

$$T \subset \{u \in \mathbb{R}^k \cap C(R(P)) : \nu \leq u_1 \leq \cdots \leq u_k\},$$

and every 1-bounded Lipschitz function f of Lipschitz constant at most 1,

$$\boxed{\sum_{\substack{n=p_1\cdots p_k \\ p_1\leq\cdots\leq p_k}} b_n f\left(\frac{\log p_1}{\log n}, \ldots, \frac{\log p_k}{\log n}\right) = \left(\sum_p b_p\right)\left(\int_T \frac{f(u)\,du}{u_1\cdots u_k} + E\right), \quad |E| \leq \frac{1}{B}.} \tag{b.2}$$

Ford–Maynard's ordinary sequence class consists of non-negative $(a_n), (b_n)$ satisfying (b.1), (b.2), with $w = a - b$ satisfying (I), (II), and (w).

8.3 The bounded class

For the bounded constants $C^{\pm}_{\mathrm{bd}}$, one additionally imposes, for some fixed $\varrho \geq 1$,

$$\boxed{|w_n| \leq \tau(n)^{\varrho} \quad (x/2 < n \leq x), \qquad \sum_p b_p \geq \frac{x}{\varrho \log x}.} \tag{4.1}$$

This extra pointwise condition is the source of the N_2 mismatch discussed later.

9 The published Ford–Maynard $1/6$ region and the class/shrinkage firewall

Theorem 9.1 (Ford–Maynard, special family)**.** *Fix $\varrho \geq 1$. For*

$$P = (1/2, 0, \nu), \qquad P_\varepsilon = (1/2-\varepsilon, \varepsilon, \nu - 2\varepsilon),$$

Ford–Maynard prove that if $\nu \geq 0.1663$, then

$$C^-(1/2, 0, \nu) > 0, \qquad \lim_{\varepsilon\to 0^+} C^-_{\mathrm{bd}}(P_\varepsilon; \varrho) > 0.$$

For the operating value $\nu_0 = 1/6$,

$$\frac{1}{6} - 0.1663 = \frac{11}{30000} > 0.$$

The theorem therefore supplies two *different* published facts:

(i) positivity for the *ordinary* class at the unshrunk point $P = (1/2, 0, 1/6)$;

(ii) epsilon-stable positivity for the *bounded* class at the shrunk points

$$P_\varepsilon = \left(\frac{1}{2} - \varepsilon,\ \varepsilon,\ \frac{1}{6} - 2\varepsilon\right).$$

It does **not** by itself assert ordinary-class positivity at P_ε. Since the 287 prime source contains von Mangoldt weights, the pointwise bounded-class condition $|w_n| \le \tau(n)^\varrho$ is unavailable. Draft V8 therefore never applies $C^-_{\mathrm{bd}}(P_\varepsilon; \varrho)$ directly to the 287 source.

The exact Type-II interval at the shrunk point is

$$\boxed{(x/2)^\varepsilon < m \le x^{1/6-\varepsilon}.} \tag{FM-window}$$

The role of the published bounded theorem is instead to identify the central Ford–Maynard geometry and a positive limiting margin. The source-specific application below replays the proof of their Theorem 8.2 and replaces the only pointwise-bounded exceptional step by an explicit N_2 estimate.

Status 9.2 (Class/shrinkage firewall)**.** There is no theorem in this dossier asserting

$$C^-(P_\varepsilon) > 0$$

for the ordinary class. Such a theorem is unnecessary if the source-specific replay in Theorem 44.1 is justified. Conversely, if the replay/localization failed, ordinary-class positivity at a shrunk point would become a genuinely new obligation.

10 Gate 0 before specialization: the actual shifted-prime source

The original programme does not begin with the 287 source $2n \pm 1$. Its Gate-0 physical sequence is a single fixed shift. We record it because a reader must be able to distinguish the programme theorem from the later 287 specialization.

For $z \ge 2$ define the odd primorial and its finite sieve density

$$P_0(z) := \prod_{\substack{p \le z \\ p \textit{ mprime},\ p>2}} p, \qquad V_0(z) := \frac{1}{2} \prod_{\substack{p \le z \\ p \textit{ mprime},\ p>2}} \left(1 - \frac{1}{p}\right). \tag{G0-1}$$

Let $I_x(n) = \mathbf{1}_{\{x/2<n\le x\}}$. The shifted-prime detector, residue-aware comparison sequence and centered difference are

$$a_x(n) := I_x(n) \frac{\Lambda(n+2)}{\log x}, \tag{G0-2}$$

$$b_{x,z}(n) := \frac{I_x(n)\, \mathbf{1}_{\{n \textit{ modd},\ (n+2, P_0(z))=1\}}}{\log x\, V_0(z)}, \qquad w_{x,z}(n) := a_x(n) - b_{x,z}(n). \tag{G0-3}$$

Both a_x and $b_{x,z}$ are non-negative and supported on $(x/2, x]$; moreover $b_{x,z}(n) = 0$ for even n. These are direct consequences of the definitions.

10.1 Finite local densities and the twin Euler product

Define

$$D_z(m) := \prod_{\substack{p \le z,\ p>2 \\ p \mid m}} \left(1 - \frac{1}{p}\right)^{-1}. \tag{G0-4}$$

Then the exact finite local-density identity is

$$\boxed{\frac{1}{2}\prod_{\substack{p\le z,\ p>2\\ p\nmid m}}\left(1-\frac{1}{p}\right)=V_0(z)D_z(m).} \tag{G0-5}$$

Indeed split the full product defining $V_0(z)$ into primes dividing m and primes not dividing m, and cancel the first group against the inverse factors in $D_z(m)$.

Consequently

$$D_z(1)=1, \tag{G0-6}$$

and for an odd prime p,

$$D_z(p)=\begin{cases}(1-1/p)^{-1}, & p\le z,\\ 1, & p>z.\end{cases} \tag{G0-7}$$

For every $a\ge 1$, $D_z(p^a)=D_z(p)$; more generally $D_z(m)$ depends only on the radical of odd m.

A second finite product is

$$W_0(z):=\prod_{\substack{p\le z\\ p>2}}\left(1-\frac{1}{p-1}\right). \tag{G0-8}$$

The elementary local identity

$$\frac{p(p-2)}{(p-1)^2}=1-\frac{1}{(p-1)^2} \tag{G0-9}$$

gives

$$\boxed{\frac{W_0(z)}{V_0(z)}=2\prod_{\substack{p\le z\\ p>2}}\frac{p(p-2)}{(p-1)^2}=2\prod_{\substack{p\le z\\ p>2}}\left(1-\frac{1}{(p-1)^2}\right).} \tag{G0-10}$$

This is an exact *finite* singular-series identity; no convergence assertion is needed for it.

10.2 The elementary pointwise comparison bound

If a Mertens lower bound

$$V_0(z)\ge\frac{c}{\log z},\qquad c>0, \tag{G0-11}$$

is available, then directly from (G0-3)

$$0\le b_{x,z}(n)\le\frac{\log z}{c\log x}. \tag{G0-12}$$

Thus pointwise boundedness of the comparison sequence is an ordinary finite-sieve consequence once (G0-11) is pinned; it is not a Type-II input.

11 Gate 0 / Gate 1 front algebra: centering, Fourier separation, CRT and the fixed determinant

This section writes out the exact finite identities which precede the later analytic estimates. They are useful because they show precisely where the expected main term disappears and where the fixed shift 2 enters.

11.1 Nonzero additive orthogonality and E-separation

Write $e_q(t) = e^{2\pi it/q}$. For $q > 0$,

$$\sum_{a \bmod q} e_q(at) = q\,\mathbf{1}_{q|t}, \tag{G01-1}$$

so after deleting the zero frequency,

$$\boxed{\sum_{\substack{a \bmod q\\ a\neq 0}} e_q(at) = q\,\mathbf{1}_{q|t} - 1,} \tag{G01-2}$$

and hence

$$\frac{1}{q}\sum_{a\neq 0} e_q(at) = \mathbf{1}_{q|t} - \frac{1}{q}. \tag{G01-3}$$

For a finite set S and coefficients c_x, put

$$C_q := \sum_{\substack{x\in S\\ q|x+2}} c_x, \qquad C := \sum_{x\in S} c_x.$$

Using (G01-3),

$$\boxed{C_q = \frac{C}{q} + \frac{1}{q}\sum_{a\neq 0} e_q(2a)\sum_{x\in S} c_x e_q(ax).} \tag{G01-4}$$

Therefore for *any* proposed expected value E,

$$C_q - E = \left(\frac{C}{q} - E\right) + \frac{1}{q}\sum_{a\neq 0} e_q(2a)\widehat{c}_q(a). \tag{G01-5}$$

The nonzero-frequency term is independent of the choice of E. Thus an incorrect source normalization cannot be repaired by changing the nonzero Fourier analysis; the zero mode and the nonzero modes are logically separate.

If $n\bar{n} \equiv 1 \pmod q$, then

$$\boxed{q \mid m + 2\bar{n} \quad\Longleftrightarrow\quad q \mid mn + 2.} \tag{G01-6}$$

This follows by multiplying the first congruence by n and using $n\bar{n} \equiv 1$.

11.2 CRT centering

For $r \geq 1$ define

$$\rho_r(y) := \mathbf{1}_{r|y} - \frac{1}{r}.$$

If $(d,p) = 1$, then $\mathbf{1}_{dp|y} = \mathbf{1}_{d|y}\mathbf{1}_{p|y}$, whence

$$\boxed{\rho_{dp}(y) = \rho_d(y)\rho_p(y) + \frac{\rho_d(y)}{p} + \frac{\rho_p(y)}{d}.} \tag{G01-7}$$

Thus a composite centered defect has a double-centered piece and two one-coordinate pieces; none may be silently removed.

At the frequency level,

$$\frac{a_d}{d} + \frac{a_p}{p} = \frac{a_d p + a_p d}{dp}, \tag{G01-8}$$

and therefore

$$\boxed{e_{dp}((a_d p + a_p d)y) = e_d(a_d y)e_p(a_p y).} \tag{G01-9}$$

This is the exact CRT frequency split underlying the four mode classes: double-centered, d-only, p-only, and zero mode.

11.3 Shift inverses and the fixed-shift determinant

Suppose

$$v = n + hq, \qquad q\bar{q} \equiv 1 \pmod{n}, \qquad v\bar{v} \equiv 1 \pmod{n}.$$

Then

$$\boxed{\bar{q} \equiv h\bar{v} \pmod{n},} \tag{G01-10}$$

so for every integer a,

$$e_n(-2a\bar{q}) = e_n(-2ah\bar{v}). \tag{G01-11}$$

The proof of (G01-10) is direct: substitute $v = n + hq$ and compare the two inverse identities modulo n.

If in addition

$$mn + 2 = qr_0, \qquad v = n + hq, \qquad s = r_0 + hm,$$

then

$$\boxed{r_0 v - ns = 2h.} \tag{G01-12}$$

Indeed $r_0(n + hq) - n(r_0 + hm) = h(qr_0 - mn) = 2h$. The determinant shift is therefore literally the original fixed shift 2, multiplied only by the translation parameter h.

11.4 Prime covariance and the exact second moment

For a finite prime set $\mathcal{P}$, define

$$K_{\mathcal{P}}(u, v) := \sum_{p\in\mathcal{P}} \rho_p(u)\rho_p(v). \tag{G01-13}$$

Expanding gives

$$\boxed{K_{\mathcal{P}}(u, v) = \sum_{\substack{p\in\mathcal{P}\\ p\mid u,\ p\mid v}} 1 - \sum_{\substack{p\in\mathcal{P}\\ p\mid u}} \frac{1}{p} - \sum_{\substack{p\in\mathcal{P}\\ p\mid v}} \frac{1}{p} + \sum_{p\in\mathcal{P}} \frac{1}{p^2}.} \tag{G01-14}$$

On the diagonal,

$$K_{\mathcal{P}}(Y, Y) = \sum_{\substack{p\in\mathcal{P}\\ p\mid Y}} \left(1 - \frac{2}{p}\right) + \sum_{p\in\mathcal{P}} \frac{1}{p^2}. \tag{G01-15}$$

For amplitudes A_i and integers Y_i, the centered form

$$F_p := \sum_i A_i \rho_p(Y_i)$$

satisfies the exact Hermitian identity

$$\boxed{\sum_{p\in\mathcal{P}} |F_p|^2 = \sum_{i,j} A_i \overline{A_j}\, K_{\mathcal{P}}(Y_i, Y_j).} \tag{G01-16}$$

No analytic saving is contained in (G01-16); it only exposes the diagonal and off-diagonal kernels exactly.

11.5 Direct Gauss reassembly

For a unit A modulo c, the congruence $Ax \equiv B \pmod{c}$ has the unique solution $x \equiv A^{-1}B$. Hence

$$\boxed{\sum_{\substack{x \bmod c \\ Ax\equiv B\ (c)}} e_c(x) = e_c(A^{-1}B).} \tag{G01-17}$$

In the direct family this produces the physical phase

$$e_c\Big(2kN\big((m+kr)p_1p_2\big)^{-1}\Big) \tag{G01-18}$$

on the unit sector. If A is not a unit, the correct solvability criterion is

$$\boxed{\exists x:\ Ax \equiv B \pmod{c} \quad\Longleftrightarrow\quad (A,c) \mid B.} \tag{G01-19}$$

Thus non-unit strata have an explicit multiplicity/gcd structure and cannot be folded into the unique-solution formula.

11.6 Nine-block regrouping and the completion exponents

For nine equal logarithmic blocks, each has mass 1/9. The balanced binary regrouping has exponents

$$U = X^{4/9+o(1)}, \qquad V = X^{5/9+o(1)}, \qquad UV = X^{1+o(1)}. \tag{G01-20}$$

At the hard modulus exponent

$$Q = X^{13/18+o(1)}, \qquad \sqrt{Q} = X^{13/36+o(1)}, \tag{G01-21}$$

one has

$$\frac{4}{9} - \frac{13}{36} = \frac{1}{12}, \qquad \frac{5}{9} - \frac{13}{36} = \frac{7}{36}, \tag{G01-22}$$

so both regrouped variables lie above the square-root modulus scale. The natural progression-count exponent is

$$\frac{4}{9} + \frac{5}{9} - \frac{13}{18} = \frac{5}{18}, \tag{G01-23}$$

and

$$\frac{13}{36} - \frac{5}{18} = \frac{1}{12}. \tag{G01-24}$$

These are exponent identities only; they do not themselves prove a square-root error term or an analytic completion theorem.

12 Gate 0: the 287 sequence dictionary

We now use the 287 sequences

$$a_X(n) = W(n/X)(\Lambda(2n-1) + \Lambda(2n+1)), \quad b_X(n) = 4W(n/X)B(n), \quad w_X = a_X - b_X.$$

Because $\operatorname{supp} W \subset (3/4, 4/5)$, these sequences are supported inside $(X/2, X]$.

12.1 The comparison prime mass (b.1) and the bounded-class prime mass

For prime p in the support,

$$B(p) = \mathfrak{S}_2 \frac{p-1}{p-2} = \mathfrak{S}_2(1 + O(1/p)).$$

The smooth prime number theorem gives

$$\sum_p b_X(p) = 4\mathfrak{S}_2 \sum_p W(p/X) + o(X/\log X) \sim 4\mathfrak{S}_2 \left(\int_0^\infty W(t)\,dt\right) \frac{X}{\log X}. \tag{G0-b1}$$

Therefore (b.1) holds for every fixed $\varpi > 1$ once X is large, and the second inequality in (4.1) holds for every fixed sufficiently large ϱ.

12.2 The ordinary growth condition (w): an explicit logarithmic proof

The comparison density has a fixed logarithmic bound, not merely an $X^{o(1)}$ bound. For every odd prime ℓ,

$$\frac{\ell-1}{\ell-2} = \frac{\ell}{\ell-1}\left(1 + \frac{1}{\ell(\ell-2)}\right).$$

Since

$$\prod_{\ell>2}\left(1 + \frac{1}{\ell(\ell-2)}\right) < \infty,$$

we obtain

$$B(n) \ll \prod_{\substack{\ell \mid n \\ \ell>2}} \frac{\ell}{\ell-1} \ll \log\log(3n), \tag{G0-Bpoint}$$

where the last estimate is the standard maximal-order/Mertens bound for $n/\varphi(n)$.

On the support $n \asymp X$,

$$a_X(n) \ll_W \log X, \qquad b_X(n) \ll_W \log\log X,$$

so

$$|w_X(n)| \ll_W \log X.$$

Using $\sum_{n\asymp X} \tau(n) \ll X \log X$ gives the explicit fixed-power logarithmic estimate

$$\boxed{\sum_n |w_X(n)|\tau(n) \ll_W X(\log X)^2.} \tag{G0-w1}$$

Thus the first part of (w) holds, for example, with any fixed $\varpi > 2$. Moreover $a_X(n) \geq 0$ and

$$w_X(n) \geq -b_X(n) \gg_W -\log\log X \geq -X^{\nu/10}$$

for every fixed $\nu > 0$ and sufficiently large X. Hence the complete ordinary condition (w) is an **elementary PASS** for the 287 sequence.

12.3 Comparison condition (b.2)

For $n = p_1 \cdots p_k$ in the region occurring in (b.2), every $p_i \geq n^\nu$, hence $k \leq 1/\nu$ and

$$B(n) = \mathfrak{S}_2 \prod_{i=1}^{k} \frac{p_i - 1}{p_i - 2} = \mathfrak{S}_2(1 + O_\nu(X^{-\nu+o(1)})).$$

Consequently (b.2) reduces to the classical fixed-k multivariate prime number theorem with a smooth radial weight. Repeated prime-number-theorem partial summation gives, for each fixed admissible T and Lipschitz f,

$$\sum_{\substack{n=p_1\cdots p_k \\ p_1\leq\cdots\leq p_k}} b_X(n) f(v(n)) = \left(\sum_p b_X(p)\right)\left(\int_T \frac{f(u)\,du}{u_1\cdots u_k} + o(1)\right). \tag{G0-b2}$$

For the Ford–Maynard definition one must make the $o(1)$ uniform over their normalized class of convex sets and Lipschitz functions; this is a standard compactness/partial-summation task but should be written with an explicit reference or proof before publication.

12.4 Type I from a weighted maximal Bombieri–Vinogradov theorem

The prime part of the Type-I sum is

$$\sum_{m\leq X^{1/2-\varepsilon}} \tau^A(m) \max_I \left| \sum_{\substack{X/2<mn\leq X \\ n\in I}} W(mn/X)\Lambda(2mn \pm 1) - \text{expected main term} \right|.$$

For fixed m, the affine prime lies in a residue class $\pm 1 \pmod{2m}$. Since $2m \leq 2X^{1/2-\varepsilon}$, a weighted maximal Bombieri–Vinogradov theorem at level $X^{1/2}(\log X)^{-C}$ has a fixed-power margin. Partial summation inserts W, and divisor-weighted versions absorb $\tau^A(m)$. The comparison term $4B(mn)$ is handled by (7.1) and Proposition 7.1 in progressions. Thus the exact Gate-0 theorem needed is the following.

Hypothesis 12.1 (287 Type-I source theorem)**.** *For every $A > 0$ and fixed $\varepsilon > 0$,*

$$\boxed{\sum_{m\leq X^{1/2-\varepsilon}} \tau^A(m) \max_I \left| \sum_{\substack{X/2<mn\leq X \\ n\in I}} w_X(mn) \right| \ll_{A,W,\varepsilon} X(\log X)^{-A}.} \tag{287-I}$$

The preceding calculation explains why this is a source-dictionary theorem at the classical 1/2 level, not the parity-breaking frontier. The literal weighted/maximal citation and the comparison-progression error remain to be pinned.

Part III

Gate 1A: the complete current conditional mechanism

13 Gate-1A parameters and canonical physical source

Let the Gate-1A ambient parameter be $X_1 \to \infty$ and set

$$M = X_1^{1/3}, \qquad R = X_1^{a}, \qquad L = X_1^{b},$$

$$H = X_1^{a+2b-2/3}, \qquad K = M/R, \qquad U = L/H, \qquad D = L^2/H.$$

Then

$$RK = M, \qquad DH = L^2, \qquad M^2H = RL^2. \tag{1A-scale}$$

The controlling vertices are

$$V_1 = (5/18, 1/3), \qquad V_2 = (5/18, 25/72), \qquad V_3 = (7/24, 1/3).$$

For a prime ℓ, put

$$\rho_\ell(n) = \mathbf{1}_{\ell|n} - \frac{1}{\ell}.$$

A canonical row is $e = (r, k, m)$ with

$$r \asymp R \text{ prime}, \qquad 1 \le |k| \le K, \qquad m \asymp M, \qquad m' = m + kr \asymp M.$$

Choose $w_0 \pmod r$ with $mw_0 \equiv -2 \pmod r$. Since $m' \equiv m \pmod r$, the same w_0 works for m'. Define

$$\alpha = \frac{mw_0 + 2}{r}, \qquad \beta = \frac{m'w_0 + 2}{r},$$

$$A_e(t) = \alpha + mt, \qquad B_e(t) = \beta + m't.$$

Lemma 13.1 (Determinant-two row identity)**.** *For every row and integer t,*

$$\boxed{m'A_e(t) - mB_e(t) = 2k.} \tag{1A-det}$$

Proof. The t-terms cancel, and

$$m'\alpha - m\beta = \frac{m'(mw_0 + 2) - m(m'w_0 + 2)}{r} = \frac{2(m' - m)}{r} = 2k.$$

$\square$

Fix one common physical weight $W_D(t) = W(t/D)$. For bounded prime coefficients b_p, d_q, define

$$P_e(t) = \sum_{p \asymp L} b_p \rho_p(A_e(t)), \qquad Q_e(t) = \sum_{q \asymp L} d_q \rho_q(B_e(t)),$$

$$C_e(b, d) = \sum_{t \in \mathbb{Z}} W_D(t) P_e(t) Q_e(t). \tag{1A-source}$$

The common-weight condition is substantive: later row dependence must be generated by the transform, not chosen independently on each row.

14 Fourier normalization and the determinant shell

The normalized pre-square coefficient has the schematic exact form

$$\widetilde{C}_{r,k,m} = \sum_{p\asymp L} b_p \sum_{\substack{q\asymp L\\ q\neq p}} d_q \sum_h \omega_{m,p,q}(h)\mathrm{e}_{mq}\left(\frac{2hkp}{m+kr}\right),$$

with

$$\omega_{m,p,q}(h) = \frac{H}{pq}\widehat{W}_D(h/pq)\mathrm{e}\left(-\frac{h\alpha}{pqm}\right). \tag{1A-omega}$$

The source norms satisfy

$$\sum_h |\omega_{m,p,q}(h)|^2 \ll H, \qquad \sum_h |\omega_{m,p,q}(h)| \ll H.$$

Additive reciprocity separates the phase, and the h-Poisson/Gauss completion produces the determinant shell

$$\boxed{ms - prn = 2.} \tag{1A-shell}$$

If $\kappa_p(m) \pmod{pr}$ satisfies $m\kappa_p(m) \equiv 2 \pmod{pr}$, then all solutions are

$$s_j = \kappa_p(m) + jpr, \qquad n_j = \frac{m\kappa_p(m)-2}{pr} + jm.$$

The quotient index has length $|j| \ll U = L/H$ and the corresponding amplitude has ℓ^2 norm $\ll UX_1^{o(1)}$.

15 The source-preserving Poisson–Bruhat input

The next statement is a genuine Gate-1A hypothesis, not a consequence of the preceding algebra.

Hypothesis 15.1 (One-sided source transport and absolute ceiling)**.** *For every fixed A_0, the transported source splits into retained states and a discarded remainder whose Gate energy is $O_{A_0}(X_1^{-A_0})$ times the natural scale. Every retained state ξ has row amplitude*

$$\boxed{c_r(\xi) = B_\xi F_\xi(r/R),} \tag{1A-PB}$$

where B_ξ is independent of r, F_ξ is supported in a fixed compact subinterval of $(1,2)$, and for every fixed J,

$$\sup_\xi \sup_t |\partial_t^J F_\xi(t)| \ll_J X_1^{o(1)}.$$

Retained states satisfy a non-negligibility cutoff relative to the Fourier truncation. The transform is one-sided with $X_1^{o(1)}$ nuclear cost and supplies

$$T_{\mathrm{abs}} := \sum_{\substack{r\asymp R\\ r\ \text{prime}}} \left(\sum_\xi |c_r(\xi)|\right)^2 \ll M^2L^4X_1^{o(1)}. \tag{1A-ceiling}$$

Finally the physical and normalized sources satisfy

$$C_e = H^{-1}\widetilde{C}_e + R_e, \qquad \sum_e |R_e|^2 \ll_{A_0} X_1^{-A_0}\frac{ML^4}{H}. \tag{1A-bridge}$$

The scale identity behind the transport budget is elementary:

$$\frac{MK(MH)^2}{RL^4} = 1. \tag{1A-budget}$$

16 Band-limited prime participation and the non-projective family energy

Put $a_{r,\xi} = |c_r(\xi)|$ and

$$M_\xi = |B_\xi| \sup_{1 \le t \le 2} |F_\xi(t)|.$$

Choose Fourier degree $N_0 = R^{1/4}$. Smoothness gives a trigonometric approximation

$$F_\xi(t) = P_\xi(t) + O_A(R^{-A}).$$

Bernstein's inequality $\|P'\|_\infty \le N_0 \|P\|_\infty$ yields an interval $I_\xi \subset [R, 2R]$ of length

$$|I_\xi| \gg R^{3/4}$$

on which $|c_r(\xi)| \gg M_\xi$.

Hypothesis 16.1 (Fixed-state exclusion)**.** *For each retained state, every row prime excluded from the generic non-projective branch divides one of $X_1^{o(1)}$ fixed nonzero integers of polynomial height.*

Huxley's prime theorem in intervals of length $x^{7/12+\epsilon}$ implies that each I_ξ contains $R^{3/4-o(1)}$ primes; deleting polynomially many fixed divisors costs only $X_1^{o(1)}$ primes. Define

$$A_\xi = \max_{r \asymp R} a_{r,\xi}, \qquad N_\xi = \frac{\sum_{r \asymp R} a_{r,\xi}}{A_\xi}.$$

Then

$$A_\xi \asymp M_\xi, \qquad N_\xi \gg R^{3/4-o(1)}. \tag{1A-part}$$

Let S be the number of prime rows, so $S = R^{1+o(1)}$. Cauchy gives the finite participation inequality

$$\left(\sum_\xi A_\xi \right)^2 \le \frac{S}{P^2} T_{\mathrm{abs}} \tag{1A-Cauchy}$$

whenever $N_\xi \ge P$ for all retained states.

Each state carries projective coordinates (Z_ξ, L_ξ) and determinant

$$\Delta(\xi, \eta) = Z_\xi L_\eta - Z_\eta L_\xi.$$

For $\Delta(\xi, \eta) \ne 0$, a row prime can couple the pair only if it divides the fixed nonzero determinant, so the pair occurs in only $X_1^{o(1)}$ rows. With $P = R^{3/4-o(1)}$, one obtains

$$\boxed{E_{\mathrm{np}} \ll R^{-1/2+o(1)} T_{\mathrm{abs}}.} \tag{1A-energy}$$

This is the genuine BPP family-energy contraction.

17 The outer square root: exact finite inequality and the source pin

The finite four-cycle inequality is

$$\sum_r \|A_r\|_{\mathrm{op}} \le |\mathcal{R}|^{1/2} \left(\sum_r \mathrm{Tr}(A_r^* A_r) \right)^{1/2}. \tag{1A-root-alg}$$

To turn this into the physical Gate contribution one still needs the following source identification.

Hypothesis 17.1 (Outer-root energy pin)**.** *Let G_{np} be the normalized Gate contribution of the non-projective transported branch. Then*

$$\boxed{G_{\mathrm{np}} \ll M^2 L^4 X_1^{o(1)} \left(\frac{E_{\mathrm{np}}}{T_{\mathrm{abs}}}\right)^{1/2}.} \tag{1A-root-pin}$$

Combining with (1A-energy) gives

$$G_{\mathrm{np}} \ll R^{-1/4+o(1)} M^2 L^4.$$

The target comparison is

$$MR^{-1/4} \le H \iff \frac{5}{4}a + 2b - 1 \ge 0.$$

At V_1, V_2, V_3 the margins are respectively

$$\boxed{\frac{1}{72}, \qquad \frac{1}{24}, \qquad \frac{1}{32}.} \tag{1A-margins}$$

The binding margin is 1/72: any later fixed-power loss must be explicitly budgeted against it.

18 The projective branch and the full state diagonal

If $\Delta(\xi, \eta) = 0$, the exact collision relation is

$$Z_\xi L_\eta = Z_\eta L_\xi.$$

For $L \neq 0$, the correct equivalence class is the ratio Z/L. For vectors $v_{Z,L} = A_Z \otimes B_L$, define

$$C_\rho = \sum_{(Z,L):Z/L=\rho} v_{Z,L}.$$

Then the finite identity is

$$\boxed{\sum_\rho \|C_\rho\|^2 = \sum_{(Z_1,L_1),(Z_2,L_2)} \mathbf{1}_{Z_1L_2=Z_2L_1} \operatorname{Re}\langle v_{Z_1,L_1}, v_{Z_2,L_2}\rangle.} \tag{1A-proj-id}$$

Since $\Delta(\xi, \xi) = 0$, the entire state diagonal is in this branch.

Hypothesis 18.1 (Projective/diagonal source routing)**.** *The physical projective source is routed into positive primitive ratio classes of multiplicity $X_1^{o(1)}$; axis and non-primitive/rank-loss states are routed to explicit exceptional blocks. Moreover*

$$\boxed{\sum_{(Z,L)} \|A_Z\|^2 \|B_L\|^2 \ll MHL^4 X_1^{o(1)},} \tag{1A-diag}$$

and every exceptional block is bounded at the same target scale.

This is a load-bearing conditional statement: it controls a branch containing the complete diagonal at the final target scale. It is not generated by the off-diagonal BPP contraction.

19 Recombination and the conditional Gate-1A theorem

Hypothesis 19.1 (Literal source recombination estimate)**.** *The quotient-cell/source recombination error satisfies*

$$\boxed{E_{\mathrm{rec}} \ll U^{-2+o(1)} M^2 L^4.} \tag{1A-rec}$$

Since $U = L/H$ and $DH = L^2$,

$$\frac{U^{-2} M^2 L^4}{M H L^4} = \frac{M}{D},$$

and the margins at V_1, V_2, V_3 are $1/18, 1/18, 1/24$.

Theorem 19.2 (Conditional canonical Gate-1A estimate)**.** *Assume Hypotheses 15.1, 16.1, 17.1, 18.1, and 19.1. Then throughout the polytope generated by V_1, V_2, V_3,*

$$\boxed{\sum_e |C_e(b,d)|^2 \ll \frac{ML^4}{H} X_1^{o(1)}.} \tag{1A-main}$$

Equivalently, in normalized form,

$$\sum_e |\widetilde{C}_e(b,d)|^2 \ll MHL^4 X_1^{o(1)}.$$

Proof. The non-projective branch follows from (1A-energy) and the outer-root pin. The complete projective branch, including the diagonal, follows from the exact ratio-class identity and Hypothesis 18.1. The source recombination error is acceptable by Hypothesis 19.1 and its exponent margin. The physical/normalized bridge in Hypothesis 15.1 gives the displayed physical estimate. □

20 What Gate 1A does and does not give

The theorem above is an analytic theorem for the canonical common-weight operator. It does not prove that every packet of an arbitrary global sieve decomposition is of this form. A source-exhaustive compiler would need a decomposition

$$S_{\mathrm{actual}} = \sum_{\lambda \in \Lambda_{\mathrm{gen}}} a_\lambda D_\lambda U_\lambda S_\lambda^\sharp + S_{\mathrm{exc}}, \qquad \sum_\lambda |a_\lambda| \|D_\lambda\|_{\mathrm{op}} \le X_1^{o(1)}, \tag{1A-global}$$

with exact packet identity, multiplicity control and exceptional routing. This is a separate theorem.

If Gate 1A is to be used for 287, a further scale dictionary is mandatory because Gate 1A uses $m \asymp X_1^{1/3}$ while the Ford–Maynard/287 Type-II variable lies in $(X/2)^\varepsilon < m \le X^{1/6-\varepsilon}$. One must specify X_1 as a function of the 287 scale, identify the physical bilinear variable, recheck R, L, H, K, U, D, and preserve the $1/72$ margin. No identity follows merely from the common letter X.

Part IV
Gate 1B: exact conductor reduction, late parent-operator reorganisation, and the pair-modulus frontier

21 Gate-1B front layer: fixed-shift determinant-2 algebra

Before the H7 conductor calculation there is a source-native determinant layer which is independently checkable. It is useful to write it explicitly because it identifies which cancellation resources really exist.

21.1 Möbius sign collapse at a distinguished prime

Suppose $q = dp$, p is prime and $(d, p) = 1$. Multiplicativity of the Möbius function gives

$$\mu(q) = \mu(d)\mu(p) = -\mu(d),$$

so

$$\boxed{\mu(d) = -\mu(q).} \tag{1B-F1}$$

On squarefree support, if $p \mid q$ and $d = q/p$, then $(d, p) = 1$ automatically. Hence all admissible distinguished prime representations of a fixed squarefree q carry the *same* Möbius sign $-\mu(q)$. There is no same-q cancellation obtained merely by summing over the distinguished prime p.

For a finite weighted cell

$$\lambda_{D,P}(q) = \sum_{\substack{dp=q \\ p\in P}} \mu(d) \log p, \qquad L_{D,P}(q) = \sum_{\substack{dp=q \\ p\in P}} \log p,$$

this gives the exact squarefree identity

$$\boxed{\lambda_{D,P}(q) = -\mu(q)L_{D,P}(q).} \tag{1B-F2}$$

21.2 Complementary divisor and determinant-2 normal form

For positive integers d, p, u, v,

$$dp \mid uv + 2 \quad \Longleftrightarrow \quad \exists \ell \geq 1 : \; uv + 2 = dp\ell. \tag{1B-F3}$$

Writing $q = dp$, the incidence is equivalently

$$\boxed{q\ell - uv = 2.} \tag{1B-F4}$$

The complementary divisor is unique and equals $(uv + 2)/q$.

Any common divisor of u and ℓ divides both uv and $q\ell$, hence divides their difference 2:

$$\boxed{(u, \ell) \mid 2.} \tag{1B-F5}$$

Similarly

$$\boxed{(v, q) \mid 2.} \tag{1B-F6}$$

so on the odd sector these pairs are coprime. This is the precise reason ordinary modular inverses become available there.

If (q_1, v_1) and (q_2, v_2) lie on the same fixed (u, ℓ) determinant-2 line,

$$q_i\ell - uv_i = 2,$$

then subtraction yields

$$\boxed{u(v_2 - v_1) = \ell(q_2 - q_1).} \tag{1B-F7}$$

Thus the solution set is an affine line; translation along the line preserves the determinant equation.

21.3 Source-native HFMV determinant identity

For two incidences

$$uv_i + 2 = d_i p_i \ell_i \qquad (i = 1, 2), \tag{1B-F8}$$

one has the exact determinant identity

$$\boxed{v_2 d_1 p_1 \ell_1 - v_1 d_2 p_2 \ell_2 = 2(v_2 - v_1).} \tag{1B-F9}$$

Proof: substitute $d_i p_i \ell_i = uv_i + 2$; the uv_1v_2 terms cancel. If $v_1 = v_2$, then

$$d_1 p_1 \ell_1 = d_2 p_2 \ell_2. \tag{1B-F10}$$

Conversely, when $u \neq 0$, equality of these products forces $v_1 = v_2$ by (1B-F8).

A finite B1 multiplicity statement also follows in the separated-prime regime: when $p_1 \neq p_2$ are primes, $|h_1|, |h_1'| \leq H$ and $2H < p_1$, the determinant congruence has at most one admissible h_1 in its interval. This supplies multiplicity one for that finite fibre. What it does *not* supply is the generic analytic off-diagonal estimate.

The source-native HFMV reduction therefore isolates the following analytic obligations:

$$\boxed{\begin{array}{c}\text{generic off-diagonal determinant variance}\\ +\ \text{diagonal divisor bound}\\ +\ \text{small proper-gcd sectors}\\ +\ \text{literal source centering}\end{array}} \tag{1B-F11}$$

The exact determinant algebra above is unconditional; the four displayed estimates are separate analytic/source theorems.

21.4 Near-top exponent geometry

In the high-modulus window $13/18 \leq \omega \leq 8/9$, write

$$Q_e = \omega, \qquad R_e = 1 - \omega, \qquad H_e = Q_e - R_e = 2\omega - 1. \tag{1B-F12}$$

Then

$$\frac{1}{9} \leq R_e \leq \frac{5}{18}, \qquad 2R_e < Q_e, \tag{1B-F13}$$

and at the binding endpoint $\omega = 13/18$,

$$H_e = \frac{4}{9}, \qquad \frac{H_e}{Q_e} = \frac{8}{13}. \tag{1B-F14}$$

The associated near-top Kloosterman exponent ledger records

$$\frac{1}{3}\frac{8}{13} - \frac{1}{5} = \frac{1}{195} > 0. \tag{1B-F15}$$

Again, (1B-F15) is an exponent margin, not by itself a Kloosterman theorem.

A useful exact local character identity is, for a nonzero field element u,

$$uv + 2 = u\left(v + 2u^{-1}\right), \tag{1B-F16}$$

hence for every multiplicative character χ,

$$\boxed{\chi(uv + 2) = \chi(u)\chi(v + 2u^{-1}).} \tag{1B-F17}$$

For odd prime modulus the map $u \mapsto 2u^{-1}$ is injective on nonzero residues. This is a finite reparametrization tool; it does not provide cancellation by itself.

22 High-order decomposition and the conductor split

The high-order source is expanded schematically as

$$a_{ubv} = \prod_{i=1}^{9}(f_i + \delta_i) = \sum_{J\subset[9]} \delta_J f_{J^c}. \tag{1B-exp}$$

An order-j term carries a determinant shell of the form

$$C_J a_J - q\ell = -2, \qquad C_J \asymp Y^j, \qquad a_J \asymp Y^{9-j}. \tag{1B-shell}$$

The exact centering, CRT, Gauss and determinant identities which feed the lower orders have been written out in Sections 11 and 21. Their role is to route the lower-complexity pieces and expose the fixed-shift determinant geometry; any analytic estimate beyond those identities must still be stated separately. Order 6 is regrouped with the order-5 geometry. H7 and H8 admit one-dimensional character/Fourier representations, while H9 is the pure-defect child.

For H7, write the induced character modulus

$$q = ce, \qquad c \asymp C, \qquad q \asymp Q.$$

The corrected coefficient-state count is

$$\sum_{c,\chi^*,e} |\beta(ce)|^2 \ll QCX^{o(1)}. \tag{1B-state}$$

The coefficient-blind estimate has relative size

$$\frac{|H_7(C)|}{Y^9} \ll \begin{cases} C/Q, & C \le Y^4, \\ C^2/(QY^4), & C \ge Y^4. \end{cases} \tag{1B-blind}$$

The formulae cross at $C = Y^4$, while the large-C estimate loses all saving at

$$\boxed{C_0 = Q^{1/2}Y^2.} \tag{1B-C0}$$

Thus C_0, not Y^4, is the high-conductor zero-saving threshold.

23 Exact high-conductor source split and spent Möbius sign

On the squarefree source,

$$\beta(q) = \sum_{dp=q} \mu(d) \log p,$$

with physical restrictions $p > V$ and $q/p > U$. In high conductor

$$e = q/c \le Y^2, \qquad V = Y^{5/2-o(1)},$$

so every distinguished source prime satisfies

$$p > V > e.$$

Hence the lane $p \mid e$ is empty. Write

$$c = pc_0, \qquad d = c_0 e,$$

and on the clean squarefree sector

$$\boxed{\beta(ce) = \mu(e)\rho_{C,e}(c), \qquad \rho_{C,e}(c) = \sum_{pc_0=c} \mu(c_0) \log p,} \tag{1B-split}$$

with the physical cell restrictions retained.

Let χ^* be primitive modulo c and let χ be its induction to ce. For squarefree $(c, e) = 1$,

$$\tau_{ce}(\chi) = \mu(e)\chi^*(e)\tau_c(\chi^*). \tag{1B-Gauss}$$

Multiplying by (1B-split),

$$\boxed{\beta(ce)\tau_{ce}(\chi) = \rho_{C,e}(c)\chi^*(e)\tau_c(\chi^*).} \tag{1B-spent}$$

Thus the short-cofactor sign $\mu(e)$ is algebraically spent and cannot be counted again as an independent cancellation resource.

24 Prime-character collapse

Write $c = pc_0$ with $(p, c_0) = 1$ and factor a primitive character as $\chi = \chi_p\chi_0$. Put

$$N = r_1 \cdots r_7 x \asymp Y^8.$$

The p-local argument has the form

$$A_p \equiv c_0 e N (2h)^{-1} \pmod p.$$

Lemma 24.1 (Prime-character collapse)**.** *For $A \in \mathbb{F}_p^\times$,*

$$\boxed{\sum_{\chi_p \ne \chi_{0,p}} \frac{\tau_p(\chi_p)}{\sqrt{p}} \chi_p(A) = \frac{p-1}{\sqrt{p}} \mathrm{e}_p(\bar{A}) + \frac{1}{\sqrt{p}}.} \tag{1B-char}$$

Proof. Summing over all multiplicative characters gives $(p-1)\mathrm{e}_p(\bar{A})$. The principal character has Gauss sum -1, so removing it adds 1. Divide by $\sqrt{p}$. □

The main term converts the reciprocal character dependence on h into the linear additive phase

$$\mathrm{e}_p(2hc_0eN). \tag{1B-linear}$$

The correction is smaller than the main local multiplier by p^{-1} and is power-negligible because $p > V = Y^{5/2-o(1)}$, provided the stated source ceiling is available.

25 Hybrid Poisson and the primitive projector

The long Fourier slot has length

$$H = Q/Y, \qquad \frac{c}{H} = \frac{Y}{e}.$$

Hence the dual h-frequency window has length $\asymp Y/e < p$.

Lemma 25.1 (Hybrid Poisson)**.** *Let $p \nmid c_0$ and let χ_0 be primitive modulo c_0. With the Fourier convention $\widehat{v}(\xi) = \int v(x)\mathrm{e}(-x\xi)\,dx$,*

$$\boxed{\sum_{h\in\mathbb{Z}} v(h/H)\chi_0(h)\mathrm{e}_p(ah) = \frac{H}{c_0}\tau_{c_0}(\chi_0) \sum_{\substack{m\in\mathbb{Z}\\ m\equiv ac_0\ (p)}} \chi_0(-m\bar{p})\widehat{v}\left(\frac{mH}{pc_0}\right).} \tag{1B-Poisson}$$

Proof. Apply Poisson summation in residue classes modulo pc_0. CRT separates the finite Fourier transform. The p-factor enforces $m \equiv ac_0 \pmod p$, while the c_0-factor is the twisted Gauss sum; the factor p from the p-sum cancels the p in the Poisson denominator. □

For $a = 2c_0eN \pmod p$, the surviving dual frequency satisfies

$$\boxed{meN \equiv 2 \pmod p, \qquad |m| \ll (Y/e)X^{o(1)} < p.} \tag{1B-dualfreq}$$

Lemma 25.2 (Primitive-character projector)**.** *For odd squarefree c_0 and $(A, c_0) = 1$,*

$$\boxed{\sideset{}{^*}\sum_{\chi_0 \pmod{c_0}} \chi_0(A) = \sum_{\substack{d|c_0\\ d|A-1}} \mu(c_0/d)\varphi(d).} \tag{1B-proj}$$

Consequently

$$\boxed{\frac{\mu(c_0)}{c_0}\sideset{}{^*}\sum_{\chi_0} \chi_0(A) = \frac{1}{c_0}\sum_{\substack{d|c_0\\ d|A-1}} \mu(d)\varphi(d).} \tag{1B-proj2}$$

Proof. This is Möbius inversion over the conductor. For squarefree c_0, $\mu(c_0)\mu(c_0/d) = \mu(d)$. □

With $A = meN/2$, the prime congruence and primitive projector combine into

$$\boxed{pd \mid meN - 2.} \tag{1B-detproj}$$

26 The H7 dual determinant and fixed-power repair

After the exact reductions, the H7 main term is

$$\begin{aligned} D_7(C) = \sum_{\substack{pc_0\asymp C\\ p>V}} \sum_{e\asymp Q/C} \log p\left(1 - \frac{1}{p}\right) \sum_{N\asymp Y^8} a_7(N) \sum_{\substack{|m|\ll(Y/e)X^{o(1)}\\ p|meN-2}} \widehat{v}(me/Y) \\ \times \frac{1}{c_0}\sum_{\substack{d|c_0\\ d|meN-2}} \mu(d)\varphi(d). \end{aligned} \tag{1B-D7}$$

Here $a_7(N)$ is the literal seven-defect plus one-model source coefficient.

Hypothesis 26.1 (H7 source norm and cell multiplicity)**.** *The physical source satisfies*

$$\sum_{N \asymp Y^8} |a_7(N)| \ll Y^8 X^{o(1)},$$

$\widehat{v}$ is rapidly decreasing with bounded supremum, the omitted dyadic/coprimality selectors have total multiplicity $X^{o(1)}$, and the primitive-character family above is the literal family generated by the source.

Proposition 26.2 (H7 fixed-power repair)**.** *Under Hypothesis 26.1,*

$$\boxed{|D_7(C)| \ll Y^9 X^{o(1)}.} \tag{1B-H7power}$$

Proof. For fixed (e, m, N), the nonzero integer $meN-2$ has only $O(1)$ prime divisors $p > V = Y^{5/2-o(1)}$. After p is fixed, write $c_0 = ds$. The absolute divisor projector is bounded by

$$\sum_{d|meN-2} \sum_{s \asymp C/(pd)} \frac{\varphi(d)}{ds} \ll X^{o(1)},$$

because the s-sum is harmonic on a dyadic interval and the divisor count is $X^{o(1)}$. Summing $e \asymp E$ and $|m| \ll Y/e$ costs $O(YX^{o(1)})$; use the ℓ^1 norm of a_7. □

Thus the previous coefficient-blind fixed-power loss Q/Y^4 has been recovered. The target, however, is

$$\boxed{D_7(C) \ll_A Y^9 (\log X)^{-A}.} \tag{H7-open}$$

This arbitrary-log saving is still open. The full divisor term $d = c_0$ reconstructs the original two-model determinant skeleton, so the character/Poisson transform is self-dual and cannot simply be iterated to closure.

27 Historical H7 natural-scale status and the late short-short promotion

The natural-scale estimate (26.2) remains an important reduction: it removed the former coefficient-blind fixed-power deficit. It is no longer the controlling final status for the entire H7 analysis. In the *short-short* exponent region

$$\boxed{\alpha < \frac{4}{9}, \qquad \beta < \frac{4}{9},} \tag{1B-SS}$$

the late Gate-1B audit isolates a joint-prime character packet. The deterministic compiler has the following conditional form.

Hypothesis 27.1 (H7 short-short common-sequence and energy inputs)**.** *After the source-exact H7 character reduction, the short-short packet is represented on one common sequence; its coefficient energy is bounded at the audited source scale; and the required multiplicative large-sieve estimate applies uniformly to that common sequence with the physical dyadic/coprimality restrictions retained.*

Proposition 27.2 (H7 short-short joint-prime capacity)**.** *Under Hypothesis 27.1, the H7 short-short contribution gains the relative fixed power*

$$\boxed{Y^{-1/2} = X^{-1/18}} \tag{1B-H7SS}$$

against the natural target scale.

The statement above is a *conditional analytic compiler*: the capacity is positive, but the source pins and common-sequence pins remain part of the theorem. It should therefore not be described as an unconditional H7 closure. The high-prime complement is disjoint from (1B-SS) and is not absorbed by this compiler.

The corresponding H8 audit reports the same fixed-power capacity after the analogous source-energy calculation. For publication purposes the status is deliberately asymmetric:

$$\boxed{\begin{array}{ll}\text{H7 short-short:} & \text{conditional compiler with } X^{-1/18} \text{ capacity,}\\ \text{H8 short-short:} & \text{same research capacity, but source/robustness interface still open.}\end{array}} \tag{1B-H78-status}$$

Thus the earlier H7 arbitrary-log determinant (H7-open) remains a valid historical source-specific target, but it is no longer the sole controlling description of the late H7 frontier.

28 The full $4/9$ complement and the square-root transition

The complement of the short-short region is organised by the exponents of the large source primes. At programme level one separates a high-d wing and a high-p wing. Joint-prime machinery has positive capacity on the high-d side and on the high-p side away from the transition where the large prime approaches $X^{1/2}$. The square-root strip is not a formal consequence of the short-short compiler and must be routed separately.

The purpose of this decomposition is logical as well as analytic: a bound in one exponent cell cannot be moved into a disjoint cell merely because both cells arise from the same order-j source. The square-root transition is therefore placed inside the shifted-quotient parent below.

29 The Full-Nine shifted-quotient parent

The high-prime switch and the pure H9 specialization can be embedded in one parent operator. Its source-exact schematic form is

$$\boxed{\mathcal{S}_{\mathrm{SQ}} = \sum_{mp \asymp X} \gamma(m)\lambda(p)\log p \; c_{\mathrm{res}}(mp-2).} \tag{1B-FullNine}$$

Here γ and λ retain the actual source-generated convolution, Möbius/sign and support structure; they are not replaced by arbitrary independent coefficients. The high-prime M-switch is one subrange of (1B-FullNine), while H9 is the pure-nine-defect specialization. Type-I or joint-large-sieve methods have capacity on strongly unbalanced sides. The balanced square-root strip is the difficult transition.

Equation (1B-FullNine) is a *parent operator*, not a theorem asserted closed. A successful estimate must control the complete source-generated coefficient family on the relevant subranges rather than a single fixed coefficient choice.

30 The QK56 full-covariance parent and its internal promotion

A second parent retains the complete covariance of the one-copy QK output. If $Q(q; h, k)$ denotes the exact one-copy output after the local transforms, the full covariance has schematic form

$$\boxed{\mathcal{C}_{\mathrm{QK}} = \sum_{q_1,q_2} \sum_{h_1,k_1,h_2,k_2} A(q_1, q_2; h_1, k_1, h_2, k_2) Q(q_1; h_1, k_1) Q(q_2; h_2, k_2).} \tag{1B-QK56}$$

The decomposition

$$\boxed{q_1 = q_2 \quad \text{versus} \quad q_1 \neq q_2} \tag{1B-QKsplit}$$

is the same-modulus diagonal versus cross-modulus off-diagonal of this one covariance parent.

30.1 Same-modulus covariance: character diagonal and cross-character Gram

For $q_1 = q_2 = q$, the source audit expands the remaining covariance into

$$\boxed{\sum_{q \asymp Q} |\beta(q)|^2 \sum_{\chi_1,\chi_2 \pmod q} \widehat{R}_q(\chi_1)\widehat{R}_q(\chi_2)W_q(\chi_1,\chi_2) = G_{\text{diag}} + G_{\text{off}}.} \tag{1B-QK-same}$$

The character diagonal $\chi_1 = \chi_2$ carries the large HK/q contribution and is routed to the existing coefficient-blind residue-energy/JQ7 estimate. The cross-character term is a centred modular-hyperbola covariance. The current internal audit estimates it through the corresponding square-root Weil/Kloosterman discrepancy and records the worst signed programme margin

$$\boxed{X^{-5/36+o(1)}.} \tag{1B-QK-margin}$$

Accordingly the programme-level status is

$$\boxed{\text{SAMEQ-QK56-CROSSCHAR-GRAM45}: \text{ internal analytic pass,}} \tag{1B-QK-same-status}$$

subject to the literal QK56/S1/S2 source transcription and a final hostile audit. This is an internal analytic claim, not a published theorem and not a Lean-closed analytic leaf.

30.2 Cross-modulus no-wrap branch: source-spread certificate

For $q_1 \neq q_2$, the late source audit uses the no-wrap regime: after the physical source parameter Θ is fixed, the residues $B_1 \pmod{q_1}$ and $B_2 \pmod{q_2}$ are fixed. The remaining product multiplicity is then measured by the push-forward of the source multiplier. The internal audit records that the relevant ℓ^2/ℓ^1 spread lies within the required source budget. We record

$$\boxed{\text{CROSSQ-THETA-SPREAD45}: \text{ internal finite/source pass,}} \tag{1B-QK-cross-status}$$

again subject to final literal transcription of the physical source.

Combining the two children gives the programme status

$$\boxed{\text{QK56-FULL-COVARIANCE45}: \text{ analytic pass modulo literal source pins.}} \tag{1B-QK-full-status}$$

This is a substantial promotion relative to Draft V9: QK56 is no longer the controlling major analytic open, provided the internal same-q/cross-q audits survive source transcription and hostile checking.

31 Canonical Gate-1B zero mode and the historical comparison fork

The canonical additive zero mode is not part of the analytic frontier. The finite/algebraic audit gives three exact statements:

(i) the canonical additive-character baseline is recovered from the standard character system;

(ii) the canonical discrepancy has zero additive Fourier frequency;

(iii) if a historical formulation retains an auxiliary comparison $E(q)$, then

$$\boxed{S_{\text{hist}} = S_{\text{can}} - R_E.} \tag{1B-zero-hist}$$

A perturbation of $E(q)$ changes R_E while leaving the canonical nonzero-frequency packet unchanged. Hence

$$\boxed{\text{canonical zero mode: algebraic PASS;} \qquad \text{historical } E(q) \text{ identification: separate source fork.}} \tag{1B-zero-status}$$

32 The balanced shifted child: recover the source before analysis

The balanced shifted-quotient branch must not be modelled as four independent arbitrary multipliers. Its source action is cyclic. Write the physical packet variables as $B_1, \ldots, B_4$ and let Θ denote the literal source map. The multiplier relation has the form

$$\boxed{a_j = \Theta(B_j, B_{j+1}) \qquad (j \bmod 4),} \tag{1B-shift-cycle}$$

and the associated character moment has schematic source form

$$\boxed{\sum_{B_1,\ldots,B_4} W(B) \sum_h W_B(h)\, \chi\Big(D\big(\Theta(B_1,B_2), \Theta(B_2,B_3), \Theta(B_3,B_4), \Theta(B_4,B_1); h\big)\Big)\,.} \tag{1B-shift-moment}$$

The four-cycle discriminant algebra is already available, but the literal pre-support-enlargement TT^* source formula has not yet been recovered. Therefore the first required action is not another black-box estimate; it is

$$\boxed{\text{SHIFT-TTSTAR-LITERAL-SOURCE-PIN.}} \tag{1B-shift-source-pin}$$

This source theorem must print, without enlargement or abstraction:

(a) the literal ranges of $B_1, \ldots, B_4$;

(b) the exact cyclic map Θ and all repeated-prime/collision constraints;

(c) the h-range, smooth weight, prefactor and normalization;

(d) the physical target before and after normalization;

(e) all zero/nonunit/shared-prime/degenerate strata and their routing;

(f) the multiplicity of every source packet under the TT^* expansion.

Only after this literal source recovery is the analytic child well posed:

$$\boxed{\text{SHIFT-SOURCE-LINKED-CHAR45.}} \tag{1B-shift-open}$$

Conditional on the QK56 promotion surviving final audit, (1B-shift-open) is the current first major analytic Gate-1B candidate.

33 Technology cards for the shifted source-linked child

The following published technologies are relevant only after the literal source has been reduced to their stated geometries.

1. Fouvry–Shparlinski estimate weighted multiplicative-character sums of the 2×2 determinant $ad - bc$ over four interval variables, with power saving once the interval length is at least $p^{1/8+\varepsilon}$. This is a serious candidate if the cyclic discriminant reduces source-exactly to determinant geometry.

2. Shparlinski's multivariate character/exponential-sum results allow monomial and reciprocal-monomial kernels with bounded one-variable weights. They become relevant if the Θ-dictionary produces such a kernel.

3. Bourgain–Garaev reciprocal sumset and multilinear Kloosterman technology is a secondary candidate after an actual reciprocal-product dictionary has been manufactured.

4. General trace-function bilinear theorems require nontrivial geometric-monodromy hypotheses. A rank-one Kummer/Legendre character is not automatically covered merely because it is a trace function, so such a theorem is not used here as a black box.

5. Blomer–Pascadi remains a fixed-Kloosterman backend-capacity card and likewise does not by itself estimate the source-linked cyclic family.

These are technology cards, not proof substitutions.

34 Historical pair-modulus umbrella and the updated controlling frontier

The earlier pair-modulus formulation remains useful as an umbrella. Both the balanced shifted TT^* child and the QK cross-modulus child lead to product moduli, and fixed-multiplier Kloosterman technology has genuine positive capacity. However the Draft-10 source split promotes the QK56 covariance internally and leaves the shifted child as the principal unresolved analytic family. Consequently the historical labels

SOURCE-QUADRATIC-MULTIPLIER45, FM-PERRON-PAIRMOD-SOURCE-MULT45

are retained as umbrella/source-grammar labels only; they are no longer the controlling first-open theorem.

35 Updated Gate-1B closure ledger

The current Gate-1B status is therefore:

Component	Current status
Orders 1–4 / low-order geometry	structural material available
H6 regroup geometry	structural material available
H7 short-short	conditional joint-prime compiler; $X^{-1/18}$ capacity
H8 short-short	same research capacity; robustness/source pin open
high-d/high-p complement	routing/capacity programme; square-root strip routed to shift parent
QK56 same-q cross-character Gram	internal analytic pass; source pins pending
QK56 cross-q source spread	internal finite/source pass; transcription pending
QK56 full covariance	analytic pass modulo literal source pins
canonical additive zero mode	exact algebraic pass
historical $E(q)$ residual	source-identification fork if retained
SHIFT-TT^* literal source pin	**SOURCE FIRST ACTION – OPEN**
SHIFT source-linked character child	**FIRST MAJOR ANALYTIC OPEN**
S1/S2 literal source pins	narrow source interfaces remain
degeneracy/collision/packet census	finite routing/reassembly remains
Gate 1B	**NOT CLOSED**

Conditional on the QK56 internal audit surviving final source transcription, the current Gate-1B picture has one principal analytic child together with literal source and finite reassembly work.

Part V

Source-exhaustive Full Type II and Gate 2

36 The F1 / F2 / F3 assembly that precedes Full Type II

The phrase "Full Type II" hides three distinct families in the programme. Their current dependency structure can be written without any formal-system terminology.

36.1 F1: global centering and the routable long-Möbius sector

Suppose the physical sequence is decomposed into finitely many pieces

$$a(n) = \sum_P A_P(n), \qquad A_P(n) = MT_P(n) + OD_P(n),$$

and define

$$b(n) := \sum_P MT_P(n), \qquad w(n) := a(n) - b(n).$$

Then identically

$$\boxed{w(n) = \sum_P OD_P(n).} \tag{F1-center}$$

This is the exact F1 global-centering identity. It does not prove that the chosen MT_P are the correct physical main terms; that is the comparison/source problem.

A long-Möbius F1 piece is declared *routable* when it contains a genuine Möbius block of mesoscopic-or-longer length and a smooth (1)-block of exponent w satisfying

$$w \geq w_*(\mu) + \delta, \qquad \delta > 0, \tag{F1-route}$$

with Möbius cancellation used exactly once before an absolute value and with no nonprincipal character placed on that Möbius block. The routing inequality is equivalent to the widened wedge

$$\boxed{122\mu + 162\theta_j < 1, \qquad \theta_j = \frac{1}{2} - w.} \tag{F1-wedge}$$

Given a routed-F3 theorem on that wedge, such an F1 piece inherits an arbitrary logarithmic saving. The missing global step is to show that *every* long-Möbius F1 piece either satisfies this route or is controlled by a separate theorem. Hence routable F1 migration is not yet full F1.

36.2 F2: prime/unit sector, principal sectors and reciprocal tensor

The F2 family splits first into a prime/unit sector and excluded strata:

$$\boxed{\begin{array}{c}\text{principal characters, nonunits, equal primes,}\\ \text{prime powers, repeated primes, gcd strata}\end{array}} \tag{F2-strata}$$

On the prime/unit sector the required transformation is a double-Mellin / multiplicative-character expansion. A complete source-exact formula for that transformation is itself still an open source theorem. Thus the statement to be proved is not replaced by a reference: one must derive the prime/unit character expansion from the physical F2 source, including principal-character subtraction, unit restrictions, repeated/equal-prime strata and all normalization factors, before the sector decomposition below is valid.

After the prime/unit reduction, the character sectors are

$$PP, PN, NP, NN.$$

The required F2 dependency is

$$\boxed{\begin{array}{c}\text{prime/unit reduction}\\ +\ PP\text{ main} + PN\text{ single-spectral} + NP\text{ single-spectral}\\ +\ NN\text{ reciprocal-character tensor}\\ +\text{ composite/gcd reassembly}\\ \Downarrow\\ \text{Full F2}\end{array}} \tag{F2-full}$$

The NN reciprocal-character tensor estimate is a genuine open analytic input. A structured tensor theorem is Ford-ready only if it simultaneously preserves the fixed shift, arbitrary divisor-bounded coefficients, rank-one structure and composite moduli. A theorem missing any of those four properties does not fill the F2 socket.

36.3 F3: fixed-depth routed pieces versus Full F3

For a fixed-depth routed F3 piece, the present interface distinguishes the high-conductor kernel from the entire piece. The needed inputs are:

$$\begin{array}{ll}(1) & \text{high-conductor kernel power saving,}\\ (2) & \text{transfer of that kernel estimate to the routed source,}\\ (3) & \text{low residual-conductor / conductor-window saving,}\\ (4) & \text{routed main-term cancellation,}\\ (5) & \text{middle-conductor large sieve.}\end{array} \tag{F3-inputs}$$

Together they give only the complete routed-piece logarithmic estimate

$$\boxed{\mathcal{P}_r \ll_A X(\log X)^{-A}.} \tag{F3-route}$$

It is important that a high-conductor kernel power saving alone does *not* imply (F3-route).

The fixed-depth route does not exhaust Full F3. In the current family decomposition the one-outer-variable piece is routed, the two-outer-variable kernel remains open, and the three-/four-outer source dictionaries remain separate obligations. Thus

$$\boxed{\text{fixed-depth routable F3} \not\Rightarrow \text{Full F3.}} \tag{F3-notfull}$$

36.4 The actual Full-Type-II assembly theorem

Let F_1, F_2, F_3 denote the completed versions of the three families above. The final project theorem has the logical shape

$$\boxed{F_1 + F_2 + F_3 + \text{source-exhaustive packet census} \Longrightarrow \text{uniform project Type II.}} \tag{F123-assemble}$$

A separate transference theorem must then identify that uniform project estimate with the Ford–Maynard Type-II predicate for the actual w_n. Only after both arrows have been proved does one obtain $\mathrm{FullFMTypeII}_{1/6}$.

37 The Full-Type-II theorem that the programme actually needs

A collection of source-specific estimates is not yet Ford–Maynard Type II. The theorem required downstream is universal in divisor-bounded coefficients and covers the full interval.

Hypothesis 37.1 (Source-exhaustive Full Type II)**.** *For the actual shifted-prime difference sequence w_n and every fixed $A > 0$, sufficiently small fixed $\varepsilon > 0$, and all divisor-bounded coefficient sequences ξ_m, κ_n,*

$$\boxed{\left| \sum_{\substack{(X/2)^{\varepsilon} < m \le X^{1/6-\varepsilon} \\ X/2 < mn \le X}} \xi_m \kappa_n w_{mn} \right| \ll_A X(\log X)^{-A}.} \tag{FullFM}$$

Moreover the global source decomposition is exhaustive: every physical packet is either an exact instance of a proved packet theorem, a controlled finite-template specialization with total operator cost $X^{o(1)}$, or an exceptional packet already below target.

Gate 1A can be one provider for some canonical packets if its five source hypotheses are proved and its scale is matched. Gate 1B is intended to provide the complementary signed/high-order packets. In addition, the F1/F2/F3 assembly in Section 36 and the source-exhaustive packet census must be completed. Neither Gate 1A nor Gate 1B, separately or by mere juxtaposition, is the universal theorem above.

38 Proof-specific Ford–Maynard Type-II landing and the source adapter

The universal hypothesis (FullFM) remains the clean published sufficient wrapper. A proof audit of Ford–Maynard Proposition 7.22 identifies a narrower proof-specific landing point. After the finite factor decomposition, the proof reaches their equation (7.23), a finite-depth product sum with one selected product in the Type-II interval. Grouping the factors gives

$$Y_1(n') = \sum_{n'=\prod_{e\in E} n_e} \prod_{e\in E} x_e(n_e), \qquad Y_2(n'') = \sum_{n''=\prod_{e\notin E} n_e} \prod_{e\notin E} x_e(n_e), \tag{FM-Y12}$$

and the universal Type-II hypothesis is invoked only after the proof has reached

$$\boxed{\sum_{n=n'n''\asymp x} Y_1(n')Y_2(n'')w_{n'n''}.} \tag{FM-7.23-gen}$$

Therefore a theorem controlling every *actual generated instance* of (FM-7.23-gen) is sufficient for this particular invocation of Type II.

Hypothesis 38.1 (FM-SIEVEGEN-TYPEII45)**.** *Every grouped bilinear sum of the form (FM-7.23-gen) actually generated by the chosen rerun of the Ford–Maynard lower-bound proof satisfies the required uniform estimate.*

This is a proof-specific sufficient target, not a Ford–Maynard theorem statement. The factor depth is fixed by the sieve parameters: in the relevant decomposition

$$\boxed{N = k\ell + r + s,} \tag{FM-depth}$$

with k, ℓ, r, s bounded in terms of the fixed sieve data. Thus an "unbounded convolution depth" obstruction is not present at this stage.

The generated one-variable factors belong to a finite-depth grammar built from Möbius factors, box/support cutoffs, Mellin/Perron twists, least/greatest-prime-factor phases, perfect-power pullbacks and bounded-depth convolutions. We retain the narrower target

$$\boxed{\text{FM-PERRON-GENERATED-TYPEII45.}} \tag{FM-PerronGen}$$

The current Ford-side archaeology records the following internal programme passes:

$$\begin{array}{ll} \text{FM-GENERATED-ONEVAR-ARCHAEOLOGY45} & \text{pass,} \\ \text{FM-FINITE-NUCLEAR-COMPILER45} & \text{pass,} \\ \text{FM-OUTER-COEFFICIENT-ABSORPTION45} & \text{pass,} \\ \text{WEIGHT-DEPENDENCE-FINITE-COMPILER45} & \text{pass.} \end{array} \tag{FM-archaeology}$$

These are programme-level source/compiler audits, not published Ford–Maynard theorems.

38.1 The unresolved Gate-output/Ford-source adapter

The next issue is not another abstract rank-reduction theorem. It is the literal source dictionary from each actual Gate packet to the generated one-variable factors entering (FM-7.23-gen). For every physical packet the adapter must provide all of the following data:

- operator and variable ranges;
- coefficient and multiplicity;
- weight law and centering;
- gcd and collision restrictions;
- original prefactor and normalized target;
- route into Gate 1A, Gate 1B, or a direct exceptional class.

We label this source-exact splice

$$\boxed{\text{SOURCE-ADAPTER45.}} \tag{SOURCE-ADAPTER45}$$

Its current status is *primary downstream source interface; source-exhaustive dictionary open.* In particular the actual weight law must be identified, not guessed: a packet must be shown to have a common W_D, a finite-template $W_{D,e}$, or a genuinely edge-dependent $W_{D,e}$.

The corresponding coordinate census is

$$\boxed{\text{FM-TO-GATE-COORDINATE-CENSUS45}: \text{ source-blocked.}} \tag{FM-Gate-census}$$

Once the literal source census exists, the finite coordinate/reassembly compiler is already available at programme level.

Hypothesis 38.2 (Ford–Maynard replacement certificate C_{FM})**.** *The source adapter and coordinate census are complete, every Type-II invocation in the chosen rerun lands in a grouped coefficient pair controlled by Hypothesis 38.1 (or the certified Perron-generated subclass), and all changes of variables, truncations, suprema, Mellin/Perron decompositions and nuclear expansions preserve the required norms uniformly.*

Only

$$\boxed{\text{SOURCE-ADAPTER45} + C_{\mathrm{FM}} + \text{FM-SIEVEGEN-TYPEII45}} \tag{FM-structured-package}$$

(or the fully certified Perron-generated variant) can replace the published universal Type-II input. Until the literal adapter and replacement certificate are constructed, the universal theorem remains the publication-safe sufficient route.

39 Ford–Maynard rough sieve weight

The published sieve construction uses a vector function g and a weight $H(n)$. In the central family, the relevant rough-support identity is the following published lemma.

Theorem 39.1 (Ford–Maynard rough-support identity, specialized notation)**.** *Let g be an admissible vector function and H the sieve weight constructed from it in the Ford–Maynard Section-7 procedure. On the relevant set $\mathcal{N}$,*

$$\boxed{H(n) = (1 \star g)(v(n))\, \mathbf{1}_{\{P^-(n) \geq n^\nu\}}, \qquad |H(n)| \ll_g 1.} \tag{FM-H}$$

If $(1 \star g)$ has fixed sign on the geometric region, then $H(n)$ has the same sign on $\mathcal{N}$.

For the epsilon-shrunk operating family set

$$\sigma := \frac{1}{6} - 2\varepsilon.$$

If $0 \leq \varepsilon \leq (1/6)/100$, then

$$\sigma \geq \frac{49}{300}.$$

Lemma 39.2 (Rough numbers have at most six prime factors)**.** *If every prime divisor of $n > 1$ is at least n^σ and $\sigma \geq 49/300$, then*

$$\boxed{\Omega(n) \leq 6,}$$

counting prime factors with multiplicity.

Proof. If $k = \Omega(n)$, then $n \geq n^{k\sigma}$, hence $k\sigma \leq 1$. Therefore $k \leq 1/\sigma \leq 300/49 < 7$. □

40 Published Ford–Maynard five-collar geometry

For

$$P_\varepsilon = \left(\frac{1}{2} - \varepsilon,\ \varepsilon,\ \nu - 2\varepsilon\right),$$

the proof of Ford–Maynard Theorem 8.3 places every exceptional vector $x \in H_2$ in a five-collar geometry: one coordinate has one of the forms

$$\boxed{\frac{1}{2} + u, \quad \nu + u, \quad 2\nu + u, \quad 1 - \nu + u, \quad 1 - 2\nu + u, \qquad |u| \le 6\varepsilon.} \tag{FM-five-collars}$$

Ford–Maynard Definition 5.1 uses

$$\boxed{v(n) = \left(\frac{\log p_1}{\log n}, \dots, \frac{\log p_k}{\log n}\right), \qquad n = p_1 \cdots p_k,} \tag{FM-vn}$$

so the collar coordinate corresponds to an individual prime factor. At $\nu_0 = 1/6$ the five centres are

$$\boxed{\frac{1}{2}, \frac{1}{6}, \frac{1}{3}, \frac{5}{6}, \frac{2}{3}.} \tag{FM-collar-centres}$$

This is a published-source geometric input. It gives a canonical way to select one collar prime in the rough exceptional sector and materially sharpens the N_2 source decomposition below.

41 Ford–Maynard Theorem 8.2: where pointwise boundedness is actually used

The bounded theorem is useful here because its proof isolates the pointwise condition to one exceptional term. Let $\mathcal{N}_1$ be the rough vectors in the main geometric region H_1 and let $\mathcal{N}_2 = \mathcal{N} \setminus \mathcal{N}_1$. The proof of Ford–Maynard Theorem 8.2 begins with

$$\begin{aligned} \sum_p w_p &\ge -\sum_p b_p + \sum_{n \in \mathcal{P} \cup \mathcal{N}_1} (w_n + b_n) H(n) \\ &= \sum_{n \in \mathcal{N}_1} b_n H(n) + \sum_{n \in \mathcal{P} \cup \mathcal{N}} w_n H(n) - \sum_{n \in \mathcal{N}_2} w_n H(n). \end{aligned} \tag{FM82-decomp}$$

The first term is evaluated from the comparison condition (b.2) through their Lemma 7.20. The second is controlled by the ordinary Type-I/Type-II/growth machinery (Proposition 7.19 and Lemma 7.21), giving

$$\sum_{n \in \mathcal{P} \cup \mathcal{N}} w_n H(n) \ll \frac{x}{(\log x)^2}. \tag{FM82-middle}$$

Only at the final term does the published proof invoke the pointwise bounded-class hypothesis:

$$n \in \mathcal{N}_2 : \qquad |w_n| \le \tau(n)^\varrho \ll_{\varrho,\nu} 1. \tag{FM82-pointwise}$$

It then converts the count of $\mathcal{N}_2$ into the geometric H_2 volume.

Thus the pointwise component of (4.1) is not needed to obtain (FM82-decomp), the comparison main term, or (FM82-middle); it is used to estimate the last $\mathcal{N}_2$ contribution. The other part of (4.1), the comparison prime-mass lower bound

$$\sum_p b_p \ge \frac{x}{\varrho \log x}, \tag{FM82-mass}$$

is still needed to normalize $x/\log x$-scale errors by $\sum_p b_p$. For the 287 comparison sequence this lower bound follows from (G0-b1) for some fixed sufficiently large ϱ.

Proposition 41.1 (Source-specific replay of Ford–Maynard Theorem 8.2)**.** *Fix $P = (\gamma, \theta, \nu)$ and a Ford–Maynard sieve function g with main/exceptional regions H_1, H_2. Assume:*

(i) the ordinary sequence conditions (b.1), (b.2), (I), (II), and (w);

(ii) for some fixed $\varrho_0 \geq 1$,

$$\sum_p b_p \geq \frac{x}{\varrho_0 \log x};$$

(iii) the source-specific exceptional estimate

$$\left| \sum_{n \in \mathcal{N}_2} w_n H(n) \right| \leq C_{N_2} \frac{x}{\log x} \sum_k \int_{H_2 \cap R_k} \frac{du}{u_1 \cdots u_k} + o\left(\sum_p b_p \right). \tag{FM82-source-N2}$$

Then the same proof gives

$$\begin{aligned} \frac{\sum_p a_p}{\sum_p b_p} \geq 1 + \sum_k \int_{H_1 \cap R_k} \frac{(1 \star g)(u)}{u_1 \cdots u_k} \, du \\ - O_g(1/B) - o(1) - C_{N_2} \varrho_0 \sum_k \int_{H_2 \cap R_k} \frac{du}{u_1 \cdots u_k}. \end{aligned} \tag{FM82-source}$$

Proof. Insert the exact decomposition (FM82-decomp). Lemma 7.20 gives the $b_n H(n)$ main integral with relative error $O_g(1/B)$. Proposition 7.19 and Lemma 7.21 give (FM82-middle), which is $o(\sum_p b_p)$ because of the prime-mass lower bound. Apply (FM82-source-N2) to the final term and use $x/\log x \leq \varrho_0 \sum_p b_p$. Dividing by $\sum_p b_p$ yields (FM82-source). No pointwise bound $|w_n| \leq \tau(n)^{\varrho}$ is used. □

42 The source-specific rough N_2 replacement via the five collars

The common normalization $a_n, b_n \mapsto a_n/\log X, b_n/\log X$ is not used as a bounded-class repair: it also shrinks the comparison prime mass by an additional logarithm. Instead we keep the physical source and use the rough geometry directly.

Set

$$\nu_0 = \frac{1}{6}, \qquad \sigma = \nu_0 - 2\varepsilon.$$

For $0 < \varepsilon \leq \nu_0/100$, Lemma 39.2 gives $\Omega(n) \leq 6$ on the σ-rough sector.

42.1 Squareful rough terms are negligible

If $p^2 \mid n \leq X$ and $p \geq X^\sigma$, then the total von-Mangoldt-weighted squareful contribution is bounded by

$$\sum_{X^\sigma \leq p \leq X^{1/2}} \left(\frac{X}{p^2} + 1 \right) \log X \ll X^{1-\sigma} \log X + X^{1/2} \log X = o\left(\frac{X}{\log X} \right). \tag{N2-squareful}$$

Thus only the squarefree rough sector contributes at the lower-bound scale.

42.2 Squarefree sector: collar prime plus largest remaining prime

By (FM-five-collars), choose canonically a collar prime q whose logarithmic coordinate lies in one of the five published collars. Let r be the largest remaining prime and write

$$\boxed{n = qdr.} \tag{N2-qdr}$$

For the shifted-prime programme source, the active two-linear-form problem is

$$\boxed{r, \qquad qdr + 2.} \tag{N2-twin-forms}$$

For the 287 two-sign specialization the corresponding source forms are

$$\boxed{r, \qquad 2qdr - 1, \qquad 2qdr + 1,} \tag{N2-287-forms}$$

with the two signs treated separately by the same upper-sieve mechanism. The collar variable is narrow, and on the rough sector the local singular factor is uniformly bounded.

Hypothesis 42.1 (Uniform collar two-linear-form sieve)**.** *Uniformly in the five collar choices, the rough prefix qd, and the residual interval for the final prime r, the relevant pair of linear forms satisfies the standard upper-bound sieve at the scale*

$$\sum_{\substack{r \in I(q,d) \\ r \text{ prime}}} \Lambda(qdr + 2) \ll \frac{|I(q,d)|}{\log X}\, \mathfrak{S}(q,d) \tag{N2-twin-sieve}$$

for the shifted-prime source, and analogously

$$\sum_{\substack{r \in I(q,d) \\ r \text{ prime}}} \Lambda(2qdr \pm 1) \ll \frac{|I(q,d)|}{\log X}\, \mathfrak{S}_{\pm}(q,d) \tag{N2-287-sieve}$$

for each 287 sign, with the local factors uniformly $O(1)$ on the rough sector.

The harmonic mass of a prime collar is

$$\sum_{q \text{ in a collar}} \frac{1}{q} \ll_{\nu_0} \varepsilon + \frac{1}{\log X}. \tag{N2-collar-mass}$$

Consequently the source audit reduces the exceptional shifted-prime lane to

$$\boxed{\sum_{n \in \mathcal{N}_2} a_n |H(n)| \ll (\varepsilon + o(1))\, \frac{X}{\log X},} \tag{N2-shifted-bound}$$

modulo Hypothesis 42.1. The comparison side is handled by the published rough-number geometry and the same five-collar volume.

Hypothesis 42.2 (Aggregate N_2 replacement)**.** *For the actual difference sequence used in the chosen application, the squareful disposal, collar decomposition, uniform upper-sieve input and comparison estimate combine to give*

$$\boxed{\left| \sum_{n \in \mathcal{N}_2} w(n) H(n) \right| \le C_{N_2} \frac{X}{\log X} \sum_k \int_{H_2 \cap R_k} \frac{du}{u_1 \cdots u_k} + o\left(\sum_p b_p \right),} \tag{N2-agg}$$

with C_{N_2} uniform on the fixed compatible epsilon range.

We record the programme label

$$\boxed{\text{FM-N2-SHIFTED-PRIME-COLLAR-SIEVE45}}$$

with status: *research pass modulo a standard uniform two-linear-form upper sieve and the literal source dictionary.* This is a sharper source decomposition than the generic coarse-cell formulation in Draft V9, but it remains external/uninhabited until the uniform sieve and physical source specialization are supplied.

43 The epsilon-compatibility gate

The shrunk parameter is not free. As $\varepsilon \downarrow 0$,

$$(X/2)^{\varepsilon} < m \leq X^{1/6-\varepsilon}$$

widens at *both* ends, so the Type-II theorem becomes harder, even though the Ford–Maynard geometric lower-bound error tends to zero. The application must therefore freeze one positive epsilon rather than pass to a limit inside the arithmetic theorem.

Let g_{FM} denote the fixed sieve function used in the published Ford–Maynard 1/6 lower-bound certificate, and write

$$\eta_{\mathrm{FM}} := 1 + \sum_{k} \int_{H(P)\cap R_k} \frac{(1 \star g_{\mathrm{FM}})(u)}{u_1 \cdots u_k}\, du > 0, \qquad P = (1/2, 0, 1/6). \tag{FM-margin}$$

Their central-family geometry gives

$$\sum_{k} \int_{H_2(P_\varepsilon)\cap R_k} \frac{du}{u_1 \cdots u_k} \ll_{g_{\mathrm{FM}}} \varepsilon. \tag{FM-H2-eps}$$

Hypothesis 43.1 (Compatible fixed epsilon)**.** *There exists one fixed*

$$\boxed{0 < \varepsilon_* \leq 1/600}$$

such that:

(i) the 287 Type-I theorem holds with exponent $1/2 - \varepsilon_*$*;*

(ii) the source-exhaustive Type-II theorem holds on $(X/2)^{\varepsilon_*} < m \leq X^{1/6-\varepsilon_*}$*;*

(iii) the rough N_2 *replacement is valid uniformly at* $\sigma = 1/6 - 2\varepsilon_*$*;*

(iv) the geometric loss in Proposition 41.1, including the constant $C_{N_2}\varrho_0$*, is* $< \eta_{\mathrm{FM}}/2$*.*

If the Type-I, Type-II and N_2 statements are proved for every sufficiently small fixed epsilon, then Hypothesis 43.1 follows by choosing one epsilon small enough after the Ford–Maynard constants are fixed. No theorem in this dossier takes $\varepsilon \to 0$ inside the arithmetic source.

44 Gate-2 conditional endgame – corrected source-specific form with two Type-II entrances

Theorem 44.1 (Conditional Gate-2 endgame)**.** *Assume:*

(i) the exact Gate-0 comparison conditions (b.1), (b.2), (w) and Type I for the physical source;

(ii) the rough-support identity and source dictionary needed to use the Ford–Maynard $H(n)$;

(iii) the source-specific aggregate N_2 replacement, Hypothesis 42.2;

(iv) the compatible fixed epsilon, Hypothesis 43.1;

(v) and one *of the following Type-II packages:*

(a) the published sufficient route: source-exhaustive Full Type II, Hypothesis 37.1; or

(b) the proof-specific route: Hypotheses 38.1 and 38.2 (or a certified narrower Perron-generated theorem together with the same replacement certificate).

Then the shifted-prime source has positive prime mass at scale X for all sufficiently large X.

Proof. Work at the single fixed point

$$P_{\varepsilon_*} = \left(\frac{1}{2} - \varepsilon_*,\ \varepsilon_*,\ \frac{1}{6} - 2\varepsilon_*\right).$$

Conditions (i)–(iv) provide the ordinary sequence class, the Ford–Maynard rough weight and the direct source-specific replacement for the final N_2 term. We do not place the unbounded von-Mangoldt source in the bounded sequence class and we do not apply C^-_{bd} directly to it.

For route (a), Hypothesis 37.1 supplies every Type-II invocation used in the ordinary Ford–Maynard machinery. For route (b), the replacement certificate C_{FM} verifies invocation-by-invocation that the structured theorem supplies the same bilinear estimates. In either case, Proposition 41.1 replays the Theorem-8.2 decomposition with the pointwise-bounded N_2 line replaced by Hypothesis 42.2.

The central Ford–Maynard geometry supplies the positive limiting margin η_{FM}, while (FM-H2-eps) and Hypothesis 42.2 control the exceptional region. By Hypothesis 43.1(iv), the total geometric/source-specific loss is less than half that margin. Taking the auxiliary parameter B and then X sufficiently large gives

$$\sum_p a_p \geq c_{\mathrm{FM}} \sum_p b_p$$

for some fixed $c_{\mathrm{FM}} > 0$. Finally (G0-b1) gives $\sum_p b_p \asymp X/\log X$. □

Status 44.2 (Downstream status after the late Gate-1B update)**.** The published universal Type-II theorem remains a sufficient wrapper. The structured route is strictly proof-specific and is *not* activated until the analytic structured estimate, SOURCE-ADAPTER45 / the FM-to-Gate coordinate census, and the complete replacement certificate C_{FM} are proved. In both routes the V8 class/shrinkage correction remains controlling: the unbounded source is handled in the ordinary class and the final pointwise-bounded N_2 step is replaced directly.

Part VI
Gold closure pivot: repaired log-cofactor lane, fixed certificate, and Kummer backend

45 Status firewall for the corrected Gold update

The fixed-certificate pivot survives, but its first packet census does not simplify as far as Draft V11 suggested. The explicit Ford certificate and the transference algebra are retained. What is withdrawn is the expectation that the literal leakage census immediately reduces to H8/H9, QK56, Gate 1A and direct packets. A source-exact smooth-parity packet occurs before that reduction.

Object	Draft-V12 status
`LOG-COFACTOR-ASYMPTOTIC287`	proved below; alternative finite route only
Current abstract `LCBeta`	specification incomplete; add $M \le 2x$ placement condition
`FIXED-CERTIFICATE287-PIN`	published/source pin PASS; explicit g_0, g_*, H_* written below
`FIXED-CERTIFICATE-TRANSFERENCE287`	proved below with the $\mathcal{N}_2$ error separated
`FINITE-H8/H9-ONLY-CENSUS`	**FAIL**; exact counterguard written below
Generic seven-prime quadratic-Kummer spectrum	conditional analytic child PASS modulo classical Weil/completion
`H8H9-SOURCE-TO-KUMMER45`	open child/source interface, but not the first fixed-leakage open
`287-FIXED-CERTIFICATE-SMOOTH-PARITY45`	**FIRST LITERAL ANALYTIC OPEN**
`287-FIXED-CERTIFICATE-LEAKAGE45`	open parent correlation (FCL)
Full generated-(7.23) census $+C_{\mathrm{FM}}$	still sufficient; fixed-g_* specialization remains relevant
Erdős #287	open

46 Repairing the log-cofactor supply interface

The broad log-cofactor blocker is a useful alternative to the fixed Sophie band, but the abstract supply predicate must place its adjacent pair inside the actual denominator span. The minimal repair is to demand

$$\boxed{M \le 2x, \qquad x+1 \le M.} \tag{LC-place}$$

Indeed Lemma 2.1 gives $N \le \lfloor M/2 \rfloor \le x$, so $x, x+1 \in [N, M]$. Consequently an abstract predicate which records only $x+1 \le M$ is not yet a literal input to an adjacent-hole theorem.

Let $C(j)$ denote the finite cofactor threshold from the finite obstruction architecture and assume the proved uniform comparison

$$C(j) \le U(j) := j\, j!. \tag{LC-C}$$

For a fixed $0 < \eta < 1/2$ put

$$J(M) := \left\lfloor \eta \frac{\log M}{\log \log M} \right\rfloor. \tag{LC-J}$$

Proposition 46.1 (Log-cofactor asymptotic)**.** *For sufficiently large M, if a prime q satisfies*

$$M \le 2J(M)q, \tag{LC-window}$$

then

$$q^2 > M, \qquad q > C(2J(M)).$$

Proof. From (LC-window),

$$q \ge \frac{M}{2J(M)} = M^{1-o(1)},$$

so $q^2 > M$ for sufficiently large M. On the other hand, by (LC-C),

$$\log C(2J) \le \log(2J) + \log((2J)!) \le \log(2J) + 2J \log(2J).$$

Since

$$J = (\eta + o(1))\frac{\log M}{\log\log M}, \qquad \log(2J) = (1 + o(1))\log\log M,$$

we obtain

$$\log C(2J) \le (2\eta + o(1))\log M.$$

Because $2\eta < 1$,

$$C(2J) \le M^{2\eta+o(1)} < \frac{M}{2J} \le q$$

for sufficiently large M. □

Thus the factorial growth of the finite cofactor threshold is not the remaining difficulty. The genuine adjacent supply theorem would still have to produce

$$\boxed{\begin{aligned} \exists x, q_0, q_1 : \quad & M \le 2x, \quad x+1 \le M, \\ & q_0 \mid x, \quad q_1 \mid x+1, \\ & q_i \text{ prime}, \quad M \le 2J(M)q_i \quad (i = 0, 1). \end{aligned}} \tag{LC-supply}$$

This simultaneous large-prime-factor correlation remains a deep alternative route; Draft V12 does not use it as the controlling analytic theorem.

47 The explicit Ford certificate and corrected fixed-certificate transference

47.1 The actual certificate g_*

Ford–Maynard's central construction is not merely existential. Write

$$\nu_0 := 0.16623, \qquad P_0 := \left(\frac{1}{2}, 0, \nu_0\right).$$

Their displayed certificate has

$$g_0(\varnothing) = 1, \qquad g_0(x) = -\mathbf{1}_{x \le 1/2},$$

$$g_0(x_1, x_2) = \mathbf{1}_{(1-\nu_0)/2 < x_1 + x_2 < 1/2},$$

$$g_0(x_1, x_2, x_3) = -\mathbf{1}_{x_1+x_2+x_3 \le 1/2},$$

and is zero in the remaining dimensions. The certificate inequality is

$$(1 \star g_0)(\mathbf{x}) \le 0 \qquad (\mathbf{x} \in \mathcal{H}(P_0)),$$

and the associated comparison constant satisfies

$$C_0 = 1 + I_3 + I_4 + I_5 + I_5' + I_6 \ge 0.000006. \tag{FC-1}$$

These are the published Ford–Maynard certificate data used here.

Fix sufficiently small $\varepsilon_* > 0$ and put

$$P_* = \left(\frac{1}{2} - \varepsilon_*,\ \varepsilon_*,\ \nu_0 - 2\varepsilon_*\right), \qquad \gamma_* := \frac{1}{2} - \varepsilon_*, \qquad \sigma_* := \nu_0 - 2\varepsilon_*.$$

Use the truncated certificate

$$g_*(\mathbf{x}) := g_0(\mathbf{x})\,\mathbf{1}_{\{|\mathbf{x}|\le 1/2-2\varepsilon_*\}}. \tag{FC-2}$$

The published shrink analysis gives

$$C_*(\varepsilon_*) = C_0 + O(\varepsilon_*). \tag{FC-3}$$

Hence there exists at least one fixed $\varepsilon_* > 0$ for which

$$\boxed{C_*(\varepsilon_*) \ge C_0/2 > 0.} \tag{FC-4}$$

This is an existence statement unless the implied constant in (FC-3) is made effective.

Status 47.1 (Fixed-certificate pin)**.** The controlling certificate is the $1/2, 0, \nu_0$ Ford–Maynard certificate above and its fixed small shrink. A numerical certificate from an unrelated parameter triple is not interchangeable with this one. We record `FIXED-CERTIFICATE287-PIN: PASS` at the published/source level, with effectiveness of a convenient decimal ε_* still separate.

47.2 Literal arithmetic weight H_*

For the canonical split $m = m_1 m_2$ satisfying

$$P^+(m_1) < n^{\sigma_*} \le P^-(m_2),$$

define

$$G_*(m;n) = \mathbf{1}_{m\le n^{\gamma_*}}\,\mu(m_1)\,g_*(\mathbf{v}(m_2;n))\,, \qquad \boxed{H_*(n) := \sum_{m|n} G_*(m;n).} \tag{FC-5}$$

This is the literal fixed-g_* specialization of the Ford–Maynard arithmetic weight.

For a prime $p \asymp X$, only $m = 1$ contributes, so

$$\boxed{H_*(p) = g_*(\varnothing) = 1.} \tag{FC-6}$$

Let $\mathcal{N}_1$ denote the main composite region where the certificate sign applies, and $\mathcal{N}_2$ the exceptional collar. Then

$$H_*(n) \le 0 \qquad (n \in \mathcal{N}_1), \tag{FC-7}$$

while $\mathcal{N}_2$ must be controlled separately by the source-specific N_2 estimate; (FC-7) is not extended to all composites.

47.3 Corrected leakage decomposition

Put

$$w_X(n) = W(n/X)\,[\Lambda(2n-1) + \Lambda(2n+1) - 4B(n)]$$

and

$$\mathcal{U}_* := (X/2, X] \setminus \big(\mathcal{P} \cup \mathcal{N}_1 \cup \mathcal{N}_2\big).$$

Define

$$\boxed{\mathcal{L}_*(X) := \sum_{n\in\mathcal{U}_*} w_X(n) H_*(n).} \tag{FC-8}$$

Also put

$$B_X := \sum_{p\in\mathcal{P}} b_X(p), \qquad \mathcal{E}_{N_2}(X) := \sum_{n\in\mathcal{N}_2} w_X(n) H_*(n).$$

Theorem 47.2 (Fixed-certificate transference). *Assume, for the fixed certificate above,*

$$\sum_{X/2<n\le X} w_X(n)H_*(n) = o(B_X), \tag{FC-9}$$

$$\mathcal{L}_*(X) = o(B_X), \tag{FC-10}$$

and

$$\mathcal{E}_{N_2}(X) = o(B_X), \tag{FC-11}$$

or more generally that the combined leakage and N_2 error are sufficiently small compared with the positive certificate margin. Assume also the comparison asymptotic

$$\sum_{n\in\mathcal{N}_1} b_X(n)H_*(n) = \big(C_*(\varepsilon_*) - 1 + o(1)\big)B_X. \tag{FC-12}$$

Then

$$\boxed{\sum_{p\in\mathcal{P}} a_X(p) \ge \big(C_*(\varepsilon_*) + o(1)\big)B_X.} \tag{FC-13}$$

In particular, if $B_X \gg X/\log X$, the affine prime mass is positive at the required scale.

Proof. By (FC-9)–(FC-11),

$$\sum_{n\in\mathcal{P}\cup\mathcal{N}_1} w_X(n)H_*(n) = o(B_X).$$

Since $a_X \ge 0$, $H_*(p) = 1$ on $\mathcal{P}$, and $H_*(n) \le 0$ on $\mathcal{N}_1$,

$$\sum_{p\in\mathcal{P}} a_X(p) \ge \sum_{n\in\mathcal{P}\cup\mathcal{N}_1} a_X(n)H_*(n).$$

Expanding $a_X = b_X + w_X$ and using (FC-12) yields (FC-13). □

For positivity one does not logically need an arbitrary logarithmic saving. A sufficient quantitative target is

$$\boxed{|\mathcal{L}_*(X)| + |\mathcal{E}_{N_2}(X)| \le \frac{C_*(\varepsilon_*)}{4}\, B_X} \tag{FC-14}$$

for all sufficiently large X, together with a correspondingly small total-correlation error. Because the available certificate margin is tiny, (FC-14) is still a strong analytic statement, but it is weaker than demanding $O_A(X(\log X)^{-A})$ for every A.

Remark 47.3 (What the pivot really saves). Fixing g_* removes the need to prove a theorem uniformly over all Ford certificates. It does *not* make the actual H_* packet family finite-dimensional in the sense needed to reduce immediately to H8/H9. The next section gives an exact counterguard.

48 The first literal fixed-certificate packet: smooth Möbius parity

The literal fixed-g_* census contains a packet before any H8/H9-only reduction. In the Ford factorisation take the cell

$$k = 0, \qquad J = \varnothing.$$

Then no prime factor exceeds the rough threshold:

$$P^+(n) \le n^{\sigma_*}.$$

Because $g_*(\varnothing) = 1$, the corresponding arithmetic weight is

$$H_*(n) = \sum_{\substack{d|n \\ d \le n^{\gamma_*}}} \mu(d) = \sum_{\substack{d|n \\ d \le n^{1/2-\varepsilon_*}}} \mu(d). \tag{SP-1}$$

Thus the fixed leakage contains the exact packet

$$\boxed{\begin{aligned} \mathcal{P}_{\mathrm{sm}}(X) := \sum_{\substack{X/2 < n \le X \\ P^+(n) \le n^{\sigma_*}}} & W(n/X)\big[\Lambda(2n-1) + \Lambda(2n+1) - 4B(n)\big] \\ & \times \sum_{\substack{d|n \\ d \le n^{1/2-\varepsilon_*}}} \mu(d). \end{aligned}} \tag{SP-2}$$

We denote this exact source packet by

$$\boxed{\texttt{287-FIXED-CERTIFICATE-SMOOTH-PARITY45.}}$$

48.1 Why the fixed census is not H8/H9-only

Consider a squarefree balanced k-prime cell

$$n = p_1 \cdots p_k, \qquad \left| \frac{\log p_i}{\log n} - \frac{1}{k} \right| < \eta,$$

where $1/k + \eta < \sigma_*$. Choose η so small that subset products of at most

$$r := \lfloor k\gamma_* \rfloor$$

primes are below n^{γ_*} and subset products of at least $r+1$ primes are above it. Then the canonical split has no large-prime component and

$$H_*(n) = \sum_{j=0}^{r} (-1)^j \binom{k}{j} = \boxed{(-1)^r \binom{k-1}{r} \neq 0.} \tag{SP-3}$$

For small fixed ε_* this gives, for example,

k	7	8	9	10	11	12
$H_*(n)$	-20	-35	70	126	-252	$-462.$

Hence H9 genuinely occurs, but so do orders $10, 11, 12, \ldots$. Therefore

$$\boxed{\texttt{FINITE-H8/H9-ONLY-CENSUS: FAIL.}} \tag{SP-4}$$

Any use of historical "Full-Nine" regrouping must prove source-exhaustiveness; the name of the parent operator does not imply that all larger $\Omega(n)$ cells have disappeared.

48.2 Parity character of the packet

At the unshrunk point $\gamma = 1/2$, for squarefree n with no exact square-root divisor, pairing $d \leftrightarrow n/d$ gives

$$\mu(n/d) = \mu(n)\mu(d).$$

When $\mu(n) = 1$, the paired full divisor sum forces cancellation below $\sqrt{n}$; when $\mu(n) = -1$, no analogous cancellation occurs. The truncated divisor sum in (SP-1) is therefore a literal divisor-parity detector. Shrinking from $1/2$ to $1/2 - \varepsilon_*$ only introduces a central-divisor collar; it does not remove this parity mechanism.

The first required analytic estimate on this packet need not be arbitrary-log. It is enough to obtain a certificate-compatible constant saving, for example

$$\boxed{|\mathcal{P}_{\mathrm{sm}}(X)| \le c_* \frac{X}{\log X}, \qquad c_* < \frac{C_*(\varepsilon_*)}{4} \times (\text{comparison constant}),} \tag{SP-5}$$

together with corresponding bounds for the other fixed-g_* packets. No provider for (SP-5) is presently identified in this dossier.

Status 48.1 (First literal open)**.** The exact smooth-parity packet (SP-2) occurs before a reduction to H8/H9 or QK56. Therefore

$$\boxed{\texttt{287-FIXED-CERTIFICATE-SMOOTH-PARITY45}}$$

is the first literal analytic open exposed by the fixed-certificate census. The parent FCL correlation remains open as well.

49 Generic quadratic-Kummer bilinear estimate

Let p be an odd prime, let χ_p be the quadratic character modulo p, and let $F \in \mathbf{F}_p[X]$ be a quadratic polynomial with nonzero discriminant. Put

$$K(x) := \chi_p(F(x)).$$

Let (α_m) be supported on an interval of length M and let (β_n) be supported on a positive interval of length $N < p$ whose elements are nonzero modulo p.

Proposition 49.1 (Quadratic-Kummer bilinear bound)**.** *With the notation above,*

$$\boxed{\left|\sum_{m,n} \alpha_m \beta_n K(mn)\right|^2 \ll p^{o(1)} \|\alpha\|_2^2 \|\beta\|_2^2 \left[M + \left(\frac{M}{\sqrt{p}} + \sqrt{p}\right) N\right].} \tag{Kum-1}$$

The only analytic input is the classical Weil bound for the complete non-square quadratic-character sum, combined with standard completion for an interval.

Proof. By Cauchy–Schwarz in m,

$$|S|^2 \le \|\alpha\|_2^2 \sum_m \left|\sum_n \beta_n K(mn)\right|^2.$$

After expansion, the inner correlation is

$$C(n_1, n_2) := \sum_{m \in I} K(mn_1)\overline{K(mn_2)}.$$

Let r_1, r_2 be the two roots of F in an algebraic closure. The polynomial $F(n_1X)F(n_2X)$ is a square only when multiplication by n_2/n_1 preserves the unordered set $\{r_1, r_2\}$; this stabilizer has at most two elements. For these exceptional ratios use the trivial bound $|C(n_1, n_2)| \le M$. For all other pairs the complete character sum is non-square, and completion plus Weil gives

$$C(n_1, n_2) \ll p^{o(1)} \left(\frac{M}{\sqrt{p}} + \sqrt{p} \right).$$

The stabilizer pairs contribute $O(M)\|\beta\|_2^2$. For the remaining pairs use

$$\left(\sum_n |\beta_n| \right)^2 \le N \|\beta\|_2^2.$$

Substitution gives (Kum-1). $\square$

Relative to the Cauchy-trivial scale $\sqrt{MN}\,\|\alpha\|_2\|\beta\|_2$, the saving factor is

$$p^{o(1)} \left(\frac{1}{N} + \frac{1}{\sqrt{p}} + \frac{\sqrt{p}}{M} \right)^{1/2}. \tag{Kum-2}$$

For a seven-prime packet use the labelled $5|2$ split

$$m = r_1 \cdots r_5 \asymp Y^5, \qquad n = r_6 r_7 \asymp Y^2.$$

The labelled prime-box multiplicity is $Y^{o(1)}$, so

$$\|\alpha\|_2^2 \ll Y^{5+o(1)}, \qquad \|\beta\|_2^2 \ll Y^{2+o(1)}.$$

If

$$Y^{5/2-o(1)} \le p \le Y^8,$$

then (Kum-2) gives

$$\left(Y^{-2} + p^{-1/2} + p^{1/2} Y^{-5} \right)^{1/2} \le Y^{-1/2+o(1)} = X^{-1/18+o(1)}$$

for $X = Y^9$.

Status 49.2 (Generic Kummer backend: conditional child provider)**.** The separated generic quadratic-Kummer spectrum has fixed-power capacity $X^{-1/18+o(1)}$ once a physical packet has been source-exactly reduced to this model. It is therefore a useful conditional analytic child provider, not a proof of the fixed-certificate leakage family. The repeated-root locus and isolated pure multiplicative-character modes remain in their separate degeneracy routers.

50 The literal Gate-1B source-to-Kummer interface

The generic estimate above does *not* by itself close H8 or H9. The remaining literal theorem is:

Hypothesis 50.1 (H8/H9 source-to-Kummer)**.** *Every actual H8/H9 packet, before source-enlarging inequalities are applied, is decomposed source-exactly into*

$$\begin{cases} \textit{a generic separated quadratic-Kummer packet covered by Proposition 49.1,} \\ \textit{a principal/zero-mode packet,} \\ \textit{a repeated-root degeneracy packet,} \\ \textit{an isolated quadratic-character packet routed separately,} \end{cases}$$

while preserving the physical prefactor, the q, p, e ranges, coefficient norms, the once-only use of each Möbius sign, labelled multiplicity, smooth/Mellin cost and source centering.

This is the exact interface denoted

$$\boxed{\texttt{H8H9-SOURCE-TO-KUMMER45.}}$$

Thus the generic character estimate is not itself an open spectral wall. The literal H8/H9 source-to-Kummer transcription remains an open Gate-1B child interface. For the fixed-certificate 287 route, however, the smooth-parity packet in Section 48 occurs earlier and is not known to reduce to this child.

51 Corrected fixed-certificate 287 route

The fixed-certificate route is logically valid but its packet family is richer than the V11 first census anticipated. Define the full leakage parent

$$\boxed{\mathcal{L}_*(X) = \sum_{n \in \mathcal{U}_*} H_*(n) W(n/X)[\Lambda(2n-1) + \Lambda(2n+1) - 4B(n)].} \tag{287-FCL}$$

We continue to denote the required parent estimate by

$$\boxed{\texttt{287-FIXED-CERTIFICATE-LEAKAGE45.}}$$

The exact smooth-parity child (SP-2) is one literal component of this parent and is presently the first exposed analytic open.

Status 51.1 (Controlling interpretation)**.** The fixed certificate removes the need for uniformity over arbitrary Ford certificates, but it does not remove the generated smooth small-prime factors, the truncated Möbius parity detector, subset-product Type-II positions, Perron/Mellin factors, or the u, v fragmentations. Consequently there is not yet evidence that the fixed-g_* route is analytically much shorter than the fixed-g_* specialization of the generated-(7.23) route.

The shortest honest dependency graph is therefore

$$
\boxed{\begin{array}{c}
\text{explicit Ford certificate } g_* + \text{fixed-certificate transference}\\
\Downarrow\\
\texttt{287-FIXED-CERTIFICATE-SMOOTH-PARITY45} + \text{other fixed-}g_*\text{ polytopal packets}\\
\Downarrow\\
\begin{cases}\text{source-exhaustive Full-Nine/QK/Gate regrouping,}\\ \text{or a genuinely new direct smooth-parity theorem}\end{cases}\\
\Downarrow\\
\texttt{287-FIXED-CERTIFICATE-LEAKAGE45}\\
+\ \text{Gate 0 / comparison / } N_2\\
\Downarrow\\
\text{positive affine prime mass}\\
\Downarrow\\
\text{287 finite blocker}\\
+\ \text{effectivity / finite bridge}\\
\Downarrow\\
\text{Erdős Problem \#287.}
\end{array}}
\tag{287-fixed-DAG}
$$

The universal Full-Type-II route and the proof-specific generated-(7.23) route remain valid sufficient alternatives. They are not mandatory merely as logical wrappers, but the fixed-g_* route has not yet demonstrated a much smaller analytic core.

Part VII

The exact 287 splice and what remains

52 The universal 287 Full-Type-II specialization (alternative sufficient route)

A clean publication-safe sufficient theorem for 287 is

$$
\boxed{\sum_{\substack{(X/2)^{\varepsilon_*}<m\le X^{1/6-\varepsilon_*}\\ X/2<mn\le X}} \xi_m\kappa_n W(mn/X)\big[\Lambda(2mn-1)+\Lambda(2mn+1)-4B(mn)\big] \ll_A X(\log X)^{-A}}
\tag{287-II}
$$

for all divisor-bounded ξ_m, κ_n. This is SOPHIE-FM-TYPEII$_{1/6}$. Draft V12 retains it as a clean sufficient alternative. The fixed-certificate route removes the universal-over-certificates wrapper, but its literal smooth-parity census still contains a genuine parity-breaking family and therefore has not been shown analytically shorter than the relevant fixed-g_* generated specialization.

A future Full-Type-II theorem can be used only after verifying all of the following, explicitly:

1. the physical affine forms are exactly $2n-1$ and $2n+1$;
2. the comparison term is exactly $4B(n)$ with the local density proved above;
3. the coefficient quantifiers are universal over divisor-bounded sequences;
4. the lower endpoint is $(X/2)^{\varepsilon_*}$ and the upper exponent is $1/6-\varepsilon_*$ at the *same fixed compatible* ε_* from Hypothesis 43.1;

5. the smooth support W is permitted without changing the theorem class;

6. no Möbius sign, Gauss factor, spectral saving, or fixed-power margin is used twice.

Gate 1A is not logically mandatory for this splice. If Full Type II is proved directly from Gate-1B plus other packets, Gate 1A may remain an auxiliary theorem. If Gate 1A is used, its scale dictionary and all five source hypotheses must be supplied.

The universal estimate (287-II) remains the clean publication-safe sufficient socket. If the proof-specific Ford–Maynard route in Section 38 is completed, one may instead seek a narrower 287 theorem for only the generated coefficient pairs used by the sieve proof. That replacement is legitimate only after a 287-specific version of C_{FM} certifies that every grouped coefficient pair produced from

$$W(n/X)\,[\Lambda(2n-1)+\Lambda(2n+1)-4B(n)]$$

belongs to the proved generated grammar. The pair-modulus Gate-1B theorem (1B-FMPM45) is a candidate provider for this narrower route, but neither the grammar certificate nor the pair-modulus theorem is currently proved.

53 From the sieve lower bound to an interior Sophie witness

Assume Gate 0, the N_2 replacement and SOPHIE-FM-TYPEII$_{1/6}$. Gate 2 yields

$$\sum_{q\ \text{prime}} W(q/X)\big(\Lambda(2q-1)+\Lambda(2q+1)\big) \gg_W \frac{X}{\log X}. \tag{287-mass}$$

Proper prime powers among $2q\pm1$ contribute only $O(X^{1/2}(\log X)^{O(1)})=o(X/\log X)$, so (287-mass) forces a genuine prime q in the support for which at least one of $2q\pm1$ is prime. Corollary 6.2 then excludes every sufficiently large counterexample.

54 Effectivity and the finite remainder

A sufficiently-large contradiction is not yet a complete solution. One must make effective every threshold used in:

- weighted maximal Bombieri–Vinogradov / progression estimates;
- the comparison-sequence asymptotics;
- the two-linear-form upper sieve and prefix-volume bounds;
- the Ford–Maynard lower-bound constant at the chosen ε;
- the smooth partitions and all $o(1)$ terms in the source compilers.

Let M_0 be the resulting computable threshold. Below M_0, it is unnecessary to enumerate all subsets. A single Sophie witness q certifies a whole interval of M satisfying the blocker inequalities. The finite certificate can therefore be a list

$$(q_i,p_i,[L_i,U_i])$$

whose intervals cover the required range, with the deeper finite obstruction bank used only on uncovered values.

55 Full theorem-by-theorem closure ledger

Statement	Status	Exact remaining obligation
Half-range placement	proved here	none
Top-layer congruence	proved here	none
Isolated-hole lemma	proved here	none
$\pm$ Sophie finite blocker	proved here	none
Fixed band $M \geq 9$ and $q > 3$	proved here	none; strict endpoint in $q^2 > M$ checked explicitly
Comparison mean $B(dm)$	proved here	none
FM conditions (I),(II),(w),(b.1),(b.2),(4.1)	published definitions	application must satisfy them literally
FM 1/6 central/bounded geometry	published theorem	ordinary unshrunk and bounded shrunk classes kept distinct
Gate-0 (b.1) / prime mass	derived here from PNT	make constants/source citation explicit
Gate-0 (w)	elementary PASS	$B(n) \ll \log\log(3n)$ gives $\sum \lvert w \rvert \tau \ll X(\log X)^2$
Gate-0 (b.2)	standard analytic reduction	write uniform multivariate-PNT proof/reference
Gate-0 Type I	source theorem needed	exact weighted maximal BV + comparison progression error
Gate-1A determinant / BPP finite mechanism	proved conditionally from displayed inputs	none in the finite algebra
Gate-1A PB transport	open source theorem	prove (1A-PB), ceiling and bridge
Gate-1A fixed-state exclusion	open source theorem	prove polynomial-height exclusion dictionary
Gate-1A outer-root pin	open source theorem	prove physical energy identification (1A-root-pin)
Gate-1A projective/diagonal bound	strong open source theorem	prove target-scale diagonal/projective bound (1A-diag)
Gate-1A recombination	open source theorem	prove literal (1A-rec)
Gate-1A canonical theorem	conditional theorem	follows when five preceding pins are supplied
Gate-1B low/medium conductor	research derivation	retain source-normalisation pins
Gate-1B source split / spent $\mu(e)$	exact algebra	none on clean squarefree sector
Gate-1B character collapse	proved here	none
Gate-1B hybrid Poisson / projector	proved here at displayed algebraic level	literal source conventions must match
Gate-1B H7 fixed-power repair	conditional proof	source norm Hypothesis 26.1

Statement	Status	Exact remaining obligation
Gate-1B historical H7 log target	historical open target	superseded as sole controlling frontier by the late short-short/source split
Gate-1B H8 short-short	research capacity / source pin open	same $X^{-1/18}$ capacity audit; robustness/source theorem still required
Full-Nine / H9 shifted parent	source-linked shifted child open	recover literal cyclic TT^* source, then prove SHIFT-SOURCE-LINKED-CHAR45
QK56 full covariance	internal analytic pass modulo source pins	same-q cross-character margin $X^{-5/36+o(1)}$; cross-q spread transcription pending
Gate-0 physical source $a_x, b_{x,z}, w_{x,z}$	written explicitly	source-to-FM normalization / analytic inputs
Gate-0 finite local density / Euler product	proved here	infinite-product and asymptotic source pins only where used
Gate-0 E-separation / CRT / shift / covariance	proved here	no analytic saving claimed
Gate-0/1 fixed determinant and Gauss reassembly	proved here	non-unit/gcd strata require separate routing
Gate-1B determinant-2 front algebra	proved here	generic HFMV analytic estimate still separate
F1 global centering	proved here	correct physical main-term identification remains source dependent
F1 routable migration	conditional	must exhaust all long-Möbius pieces
F2 full family	open	PP/PN/NP/NN plus composite/gcd reassembly; NN tensor open
F3 routed fixed-depth sector	conditional / partial	two-outer kernel and remaining source families prevent Full F3
Full F1/F2/F3 assembly	open	prove (F123-assemble) and source-exhaustive census
Source-exhaustive universal Full Type II	open sufficient wrapper	prove Hypothesis 37.1 plus packet census
FM-SIEVEGEN-TYPEII45	proof-specific open interface	prove Hypothesis 38.1 for every actual generated (7.23) instance
FM-PERRON-GENERATED-TYPEII45	proof-specific sufficient target; grammar/source certificate open	fixed-depth grammar archaeology passed internally
SOURCE-ADAPTER45 / FM-to-Gate coordinate census	primary source interface open	print every physical packet, ranges, weight law, multiplicity, prefactor, target and route
Ford–Maynard replacement certificate C_{FM}	open finite/source theorem	after the source adapter, certify every generated Type-II invocation and transformation
FM rough-support identity	published theorem	exact source instantiation
$\Omega(n) \le 6$	proved here	none
FM five-collar geometry	published-source pass	no new geometric theorem required
N_2 squareful rough lane	elementary pass	negligible by (N2-squareful)

Statement	Status	Exact remaining obligation
N_2 collar two-form lane	research pass modulo standard upper sieve	prove Hypothesis 42.1 uniformly for the physical source
Aggregate N_2 replacement	source-specific open splice	instantiate collar sieve + comparison side uniformly at fixed ε_*
Source-specific FM-8.2 replay	conditional theorem proved here	ordinary FM machinery + direct N_2 replacement; no direct use of C_{bd}^-
Compatible ε_*	open compatibility gate	prove Hypothesis 43.1 at one fixed positive epsilon
Gate-2 endgame	corrected conditional theorem	Gate 0 + rough N_2 + compatible fixed epsilon + either universal Full Type II or certified proof-specific replacement
Repaired log-cofactor placement	proved here	abstract supply must include $M \leq 2x$ as well as $x+1 \leq M$
Log-cofactor threshold asymptotic	proved here	only the simultaneous adjacent large-factor supply remains open on that route
Explicit fixed certificate g_*	published/source PASS	effectiveness of a convenient decimal ε_* remains separate
Fixed-certificate transference	proved here	total correlation + leakage + N_2 error + positive comparison margin
Smooth-parity fixed-certificate packet	FIRST LITERAL ANALYTIC OPEN	prove a certificate-compatible saving for (SP-2)
Finite H8/H9-only census	FAIL	higher balanced defect orders occur literally by (SP-3)
Generic quadratic-Kummer bilinear backend	conditional child provider	gives $X^{-1/18+o(1)}$ only after source-exact reduction
H8/H9 source-to-Kummer	open child/source theorem	relevant to H8/H9 packets but not exhaustive for fixed leakage
287 fixed-certificate leakage	open parent correlation	bound all fixed-g_* packets, including smooth parity
All-generated-(7.23) census $+C_{\mathrm{FM}}$	alternative sufficient route	fixed-g_* specialization may still be needed to treat smooth small-prime packets
287 universal Type-II specialization	open alternative sufficient route	prove (287-II) only if using the universal/full-Type-II path
Effectivity	open	compute all thresholds
Finite remainder	finite task	interval-cover certificates below M_0

56 End-to-end dependency graph

$$\begin{array}{c}
\textbf{Corrected fixed-certificate 287 route} \\
\text{explicit Ford certificate } g_* + \text{fixed-certificate transference} \\
\Downarrow \\
\texttt{287-FIXED-CERTIFICATE-SMOOTH-PARITY45} + \text{other fixed-}g_* \text{ packets} \\
\Downarrow \\
\texttt{287-FIXED-CERTIFICATE-LEAKAGE45} + \text{Gate 0/comparison/}N_2 \\
\Downarrow \\
\text{positive affine prime mass} \\
\Downarrow \\
\text{interior Sophie witness} \\
\Downarrow \\
\text{287 finite blocker} \\
\Downarrow \\
\text{no sufficiently-large counterexample} \\
+ \text{ effectivity} + \text{finite certificates} \\
\Downarrow \\
\textbf{Erdős Problem \#287.}
\end{array}$$

The Gate-1B generic high-order backend now has the separate internal chain

$$\begin{array}{c}
\text{actual H8/H9 source} \\
\Downarrow \ \texttt{H8H9-SOURCE-TO-KUMMER45} \\
\text{generic quadratic-Kummer packet} \\
\Downarrow \ \text{Proposition 49.1} \\
X^{-1/18+o(1)} \text{ saving.}
\end{array}$$

plus the explicit zero/principal and degeneracy routers. This is relevant both to the Twin-Prime programme and to any fixed-certificate leakage decomposition that produces those packets.

Alternative sufficient routes. The longer programme remains mathematically valid and important:

$$\begin{array}{c}
\text{Gate 0} + \text{Gate 1A} + \text{Gate 1B} + \text{F1/F2/F3} \\
+ \ \texttt{SOURCE-ADAPTER45} \\
\Downarrow \\
\text{FullFMTypeII}_{1/6} \\
\text{or a certified } C_{\mathrm{FM}} + \text{FM-SIEVEGEN-TYPEII45.}
\end{array}$$

followed by the corrected rough-N_2 Gate-2 replay. Draft V12 no longer treats completion of this whole wrapper as logically mandatory for Problem #287 once the fixed-certificate route is available.

57 What this document claims and does not claim

The finite unit-fraction part through Corollary 6.2 is written as a complete mathematical proof. The comparison mean is also proved. The Ford–Maynard theorem statements are external published inputs. Gate 0 is written from its physical shifted-prime detector through local densities, centering, CRT, fixed-shift determinants and the exact Ford–Maynard sequence conditions. Gate 1A is a conditional theorem whose complete current mechanism and five load-bearing source hypotheses are displayed. Gate 1B includes the determinant-2 and HFMV front algebra, the historical H7 dual-determinant repair, the late H7 short-short $X^{-1/18}$ conditional compiler, the H8 capacity audit, the full 4/9 complement routing, the Full-Nine shifted-quotient parent, the QK56 covariance parent, the canonical zero-mode pass and the moving pair-modulus source-multiplier frontier. The F1/F2/F3 assembly preceding universal Full Type II is written out separately, and the later Ford–Maynard proof audit adds a second, proof-specific structured Type-II entrance which remains conditional on the literal SOURCE-ADAPTER45 packet dictionary and a complete replacement certificate. Gate 2 is a corrected conditional theorem: the dossier replays the Ford–Maynard Theorem-8.2 decomposition for the ordinary class and replaces the pointwise-bounded N_2 step by an explicit source-specific open lemma. It does not apply bounded-class positivity directly to the unbounded 287 source. Accordingly this dossier is designed to make the remaining mathematical work auditable; it is not a proof that the remaining hypotheses hold.

Draft V12 additionally pins the explicit Ford certificate, writes its literal arithmetic weight, proves the corrected fixed-certificate transference identity with the N_2 error separated, and exhibits an exact smooth-parity child of the leakage family. Balanced squarefree cells prove that the fixed census is not H8/H9-only. The generic quadratic-Kummer backend with $X^{-1/18+o(1)}$ capacity is retained only as a conditional child provider. The document does not claim the H8/H9 source-to-Kummer transcription, the smooth-parity estimate, the full fixed-certificate leakage estimate, or Problem #287.

58 Draft V12 corrected Gold-pivot delta

Relative to Draft V11, the controlling corrections are:

1. The fixed Ford certificate is no longer abstract: the dossier records the explicit central g_0, the shrunk fixed g_*, the positive certificate constant $C_*(\varepsilon_*)$, and the literal arithmetic weight H_*.

2. The transference theorem is corrected to separate the main composite region $\mathcal{N}_1$ from the exceptional collar $\mathcal{N}_2$. Total correlation, fixed leakage and the N_2 source error are distinct antecedents.

3. Arbitrary-log leakage saving is no longer presented as logically necessary. A sufficiently strong certificate-margin constant saving is enough for positivity.

4. The expectation
$$\text{fixed leakage} \longrightarrow \{\text{H8/H9}, \text{QK56}, \text{1A}, \text{direct}\}$$
is withdrawn. The exact $k = 0$, $J = \varnothing$ cell gives the smooth-parity packet (SP-2).

5. Balanced squarefree k-prime cells give
$$H_*(n) = (-1)^r \binom{k-1}{r} \neq 0, \qquad r = \lfloor k\gamma_* \rfloor,$$
so orders $10, 11, 12, \ldots$ occur literally. Hence `FINITE-H8/H9-ONLY-CENSUS: FAIL`.

6. `287-FIXED-CERTIFICATE-SMOOTH-PARITY45` is the first literal analytic open exposed by the fixed-certificate census. The parent `287-FIXED-CERTIFICATE-LEAKAGE45` remains open.

7. The seven-prime Kummer estimate remains useful with $X^{-1/18+o(1)}$ capacity, but only as a conditional analytic provider after source-exact reduction. It does not close the smooth-parity packet.

8. The fixed-certificate pivot remains logically valid, but the dossier no longer claims evidence that it is analytically much shorter than the fixed-g_* generated-(7.23) route.

9. All prior safeguards remain in force: the repaired log-cofactor placement, ordinary/bounded Ford–Maynard firewall, source-specific N_2 replacement, fixed compatible ε_*, Gate-1A source pins, Gate-1B provenance, effectivity and finite completion.

References

[1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L'Enseignement Mathématique, 1980, p. 33.

[2] Various authors, *Some of Paul's Favorite Problems*, booklet for the conference "Paul Erdős and his mathematics", Budapest, July 1999, Problem 1.15.

[3] K. Ford and J. Maynard, *On the Theory of Prime-Producing Sieves*, arXiv:2407.14368 (2024).

[4] M. N. Huxley, *On the Difference between Consecutive Primes*, Invent. Math. 15 (1971/72), 164–170.

[5] A. Zygmund, *Trigonometric Series*, 3rd ed., Cambridge University Press, 2002.

[6] S. Bettin and V. Chandee, *Trilinear Forms with Kloosterman Fractions*, Adv. Math. 328 (2018), 1234–1262.

[7] T. Wright, *Trilinear Kloosterman Fractions I: Partially Fixed Moduli and Unbalanced Convolutions*, arXiv:2604.25177 (2026).

[8] V. Blomer and A. Pascadi, *Bilinear Forms with Kloosterman Sums via Quadratic Characters*, arXiv:2607.24311 (2026).

[9] A. Pascadi, *Non-Abelian Amplification and Bilinear Forms with Kloosterman Sums*, arXiv:2511.08445 (2025).

[10] E. Fouvry and I. E. Shparlinski, *On Character Sums with Determinants*, arXiv:2210.15761 (2022).

[11] I. E. Shparlinski, *Multiple Exponential and Character Sums with Monomials*, arXiv:1309.0324 (2013).

[12] J. Bourgain and M. Z. Garaev, *Sumsets of Reciprocals in Prime Fields and Multilinear Kloosterman Sums*, arXiv:1211.4184 (2012).

[13] E. Fouvry, E. Kowalski, P. Michel and W. Sawin, *Bilinear Forms with Trace Functions*, arXiv:2511.09459 (2025).